

Rebuilding the SMP Branch from xpack-development

Comprehensive Architectural Analysis, Step-by-Step Implementation Runbook & GitHub Pull Request Blueprint

1. Executive Summary & The GitHub PR Paradigm

1.1 The Genesis & Pitfalls of Monolithic SMP Development

When the multi-core Symmetric Multiprocessing (smp) branch was originally developed for `μOS++ IIIe`, it was implemented as a monolithic, sweeping conversion on top of `xpack-development`. In that initial push, hundreds of disparate changes were introduced simultaneously across multiple git repositories:

1. **Unrelated Single-Core Defect Fixes** (POSIX `dirent.h` ISO C empty struct, double-list iterator semantics, newlib syscall types, memory allocator usable size, steady clock duration 64-bit overflow) were interleaved with multi-core scheduler commits.
2. **Architectural Restructuring** (dissolution of `micro-os-plus-iii-devices`, migrating SoC headers into architecture ports) was commingled with thread management state machines.
3. **Silicon Extensions** (ARM Cortex-M33, Raspberry Pi RP2350, AArch32, and AArch64 ports) were added in large batches alongside test harness modifications.

While the resulting smp branch in `~/Work2/micro-os-plus/` achieved full multi-core functionality and passing tests, its commit history is **difficult to review, bisect, cherry-pick, or upstream via standard GitHub Pull Requests**. Upstream maintainers cannot easily audit a 10,000-line diff spanning 7 repositories to verify that uniprocessor behavior remains pristine.

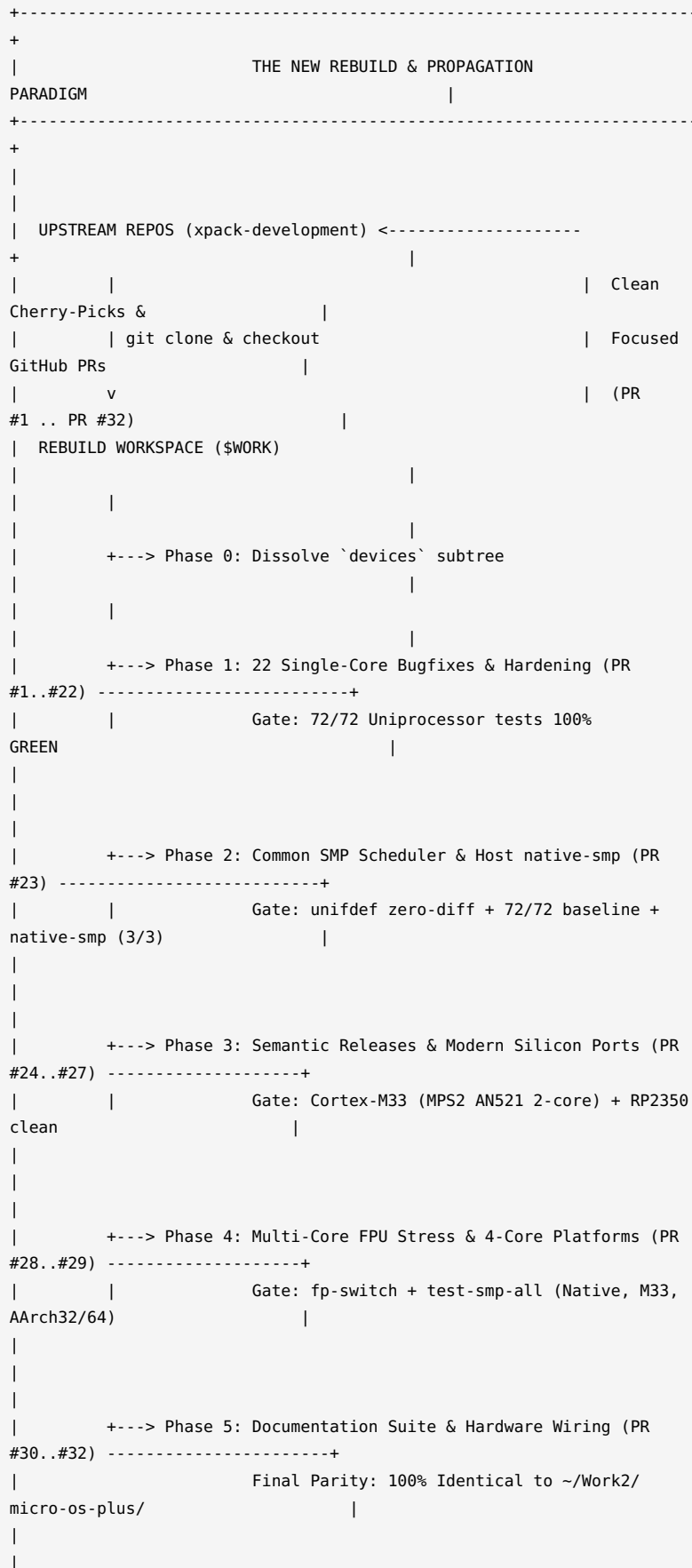
1.2 The New Paradigm: Small, Focused, Logically Related Commits

This document establishes the **New SMP Development Procedure**. Under this methodology, the smp branch is rebuilt from scratch, starting directly from clean upstream `origin/xpack-development` branches across all repositories.

Every step in this rebuilt history adheres strictly to the **GitHub Pull Request paradigm**:

- **Single-Subject Focus:** Every commit addresses exactly **one** issue, defect, feature, or architectural refactoring. Unrelated changes are never commingled.
- **Independent Reviewability:** Each commit contains its own clear rationale, problem description, failure mode averted, and unified diff that can be reviewed in isolation.

- **Bi-directional Propagation:** Commits developed on the new smp branch can be cleanly cherry-picked or opened as isolated Pull Requests targeting xpack-development.
- **Test Framework Discipline:** Every commit compiles cleanly and passes the rigorous µOS++ test framework gate (72/72 baseline uniprocessor tests, followed by multi-core suites as capabilities land).



+-----
+

Procedure update (Part II, §11-§18): The rebuild is carried out **only with git commits**. No bash or Python scripts are used: no `scripts/smp/`, no chunk recipes, no stashes and no `git checkout golden -- <files>` lifts. Each commit only modifies or adds files, covers one subject (one fix, one feature, one test, one board, one document), and becomes one GitHub pull request. §11-§18 are the authoritative per-repository commit lists for `micro-os-plus-iii`, `cortexm`, `posix-arch`, `aarch32` and `aarch64`, including the retirement of devices. Where §3, §9 and §10 disagree with them, §11-§18 apply.

Execution record (Part III, §20-§22): Part II was executed on 2026-10-06. §20 summarizes every step from `xpack-development` (single CPU) to `smp` (SMP) across all repositories. §21 records the deviations from the plan, the builds, the parity check and the final test results. §22 lists every commit of every repository (kernel, `posix-arch`, `cortexm`, `aarch32`, `aarch64`, devices) as a pull request: number, head and base branch, SHA, category, stage/theme and subject, plus the cross-repository merge order.

1.3 Target Architecture: The Five Clean Repositories & Dissolution of devices

In the original architecture, device-specific drivers and SoC registers were segregated into a separate `micro-os-plus-iii-devices` repository. This model introduced unnecessary cross-repository coupling and circular dependencies during multi-core bringup.

In the new architecture-specific model, `micro-os-plus-iii-devices` is **dissolved**. Its SoC and driver contents are permanently folded into their owning architecture repositories:

<code>micro-os-plus-iii/</code>	(Kernel, portable POSIX I/O, core test framework)
<code>micro-os-plus-iii-cortexm/</code>	(ARMv7-M & ARMv8-M Cortex-M architectures + STM32 / RP2350 SoCs)
<code>micro-os-plus-iii-aarch32/</code>	(ARMv7-A 32-bit architecture + BCM2837 / Raspberry Pi SoCs)
<code>micro-os-plus-iii-aarch64/</code>	(ARMv8-A 64-bit architecture + BCM2837 / Raspberry Pi SoCs)
<code>micro-os-plus-iii-posix-arch/</code>	(Host POSIX architecture + native uniprocessor & native-smp ports)

Compatibility shims (`micro-os-plus::devices` CMake ALIAS targets) ensure that existing external projects referencing the legacy target continue to configure and link without modification.

Crucial Architecture Decision: Following the dissolution of its contents into the respective architecture repositories, the `micro-os-plus-iii-devices` Git repository and folder are **permanently deleted and eliminated** from the active workspace. The final ecosystem consists strictly of the five primary architecture and kernel repositories.

2. Repository Architecture & Workspace Bootstrap Procedure

2.1 Workspace Directory Layout

Rebuilding the SMP branch cleanly requires an isolated workspace folder (\$WORK), distinct from the reference workspace in ~/Work2/micro-os-plus/.

```
# Recommended workspace setup
export WORK="/home/dan/Work2/micro-os-plus-rebuild"
mkdir -p "$WORK" && cd "$WORK"
```

The workspace contains the five primary active repositories, plus the legacy devices repository used strictly during Phase 0:

Directory	Repository Name	Initial Branch	Role in Rebuild
\$WORK/micro-os-plus-iii	micro-os-plus-iii	xpack-development	Kernel, schedulers, POSIX I/O, test suites
\$WORK/micro-os-plus-iii-cortexm	micro-os-plus-iii-cortexm	xpack-development	Cortex-M0/M3/M4/M7 baseline + M33/RP2350 SMP
\$WORK/micro-os-plus-iii-posix-arch	micro-os-plus-iii-posix-arch	xpack-development	Host POSIX uniprocessor + dual-core native-smp
\$WORK/micro-os-plus-iii-aarch32	micro-os-plus-iii-aarch32	xpack-development	ARMv7-A 32-bit Cortex-A multi-core port
\$WORK/micro-os-plus-iii-aarch64	micro-os-plus-iii-aarch64	xpack-development	ARMv8-A 64-bit Cortex-A multi-core port
\$WORK/micro-os-plus-iii-devices	micro-os-plus-iii-devices	smp / master	Temporary source for Phase 0 subtree migration (permanently deleted after dissolution)

2.2 Cloning Repositories & Branch Topology

All repositories are cloned from their canonical remotes. The clean baseline is checked out, and a dedicated rebuild branch (named smp-rebuild or smp) is created:

```
cd "$WORK"

# 1. Clone the core repositories
git clone https://github.com/micro-os-plus/micro-os-plus-iii.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-cortexm.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-posix-arch.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-aarch32.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-aarch64.git
git clone https://github.com/micro-os-plus/micro-os-plus-iii-devices.git

# 2. Add local reference remotes pointing to golden source in ~/Work2/micro-os-plus/
for repo in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-iii-posix-arch \
            micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64; do
```

```
cd "$WORK/$repo"
git remote add golden "/home/dan/Work2/micro-os-plus/$repo"
git fetch golden
git checkout -b smp origin/xpack-development
done
```

2.3 Toolchain Verification (check-env.sh)

Before applying changes, verify that the environment has all required native and cross-compilers, debuggers, QEMU emulators, and build utilities:

```
cd "$WORK/micro-os-plus-iii"
# Execute environment checker
bash scripts/smp/check-env.sh
```

Required toolchain components:

- **Build Tools:** CMake ≥ 3.20 \ge 3.20, Ninja ≥ 1.10 \ge 1.10, xPacks Package Manager (xpm) ≥ 0.14 \ge 0.14.
- **Host Compilers:** GCC 11, 12, 13, or 14; Clang 16, 17, 18, or 19.
- **Cross Toolchains:** arm-none-eabi-gcc ≥ 12.3 \ge 12.3, aarch64-none-elf-gcc ≥ 12.3 \ge 12.3.
- **Emulators:** qemu-system-arm ≥ 7.0 \ge 7.0 (supporting mps2-an521, raspi3b), qemu-system-aarch64 ≥ 7.0 \ge 7.0.
- **Utilities:** unifdef (for single-core projection verification), Python 3 with pypdf, weasyprint, and markdown.

2.4 Development Linking Protocol (xpm link)

In the $\mu\text{OS++}$ development workflow, repositories locate each other via xpm link. This ensures that local edits in architecture ports are immediately picked up when compiling kernel test suites:

```
# Register port development links
cd "$WORK/micro-os-plus-iii-posix-arch" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-cortexm" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-aarch32" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-aarch64" && xpm install && xpm link

# Link into the kernel test harness
cd "$WORK/micro-os-plus-iii"
xpm link @micro-os-plus/cortexm
xpm link @micro-os-plus/posix-arch
```

2.5 The Test Framework Paradigm of micro-os-plus-iii

The test paradigm in $\mu\text{OS++}$ IIIe is strictly structured:

1. **The 72-Test Baseline Matrix:** Comprises 24 distinct build configurations (combining native Linux GCC/Clang with Cortex-M0, M3, M4, M7 cross-compilations) each executing three standardized suites:
 - rtos-apis-test: Comprehensive functional validation of all POSIX and RTOS primitives.
 - mutex-stress-test: High-contention lock/unlock concurrency verification.
 - cmsis-os-validator-test: ARM official CMSIS-RTOS validation suite (60/60 test cases).
 - **Baseline Invariant:** At every commit in Phase 1, xpm run test-all must produce 72/72 PASS (100% green).

2. The Multi-Core SMP Test Matrix (test-smp-all):

- test-native-smp: Real dual-core host execution using pthreads-as-cores and signal-based IPIs.
 - test-2xcortex-m33-cmake: Dual-core ARMv8-M execution on QEMU mps2-an521.
 - test-aarch32-rpi3b: 4-core ARMv7-A execution on QEMU raspib3.
 - test-aarch64-rpi3b: 4-core ARMv8-A execution on QEMU raspib3.
 - fp-switch: Floating-point register corruption stress test under aggressive preemption.
-

3. Phase 0: Pre-Migration Subtree Dissolution of Legacy devices

3.1 Why the devices Repository Must Disappear

The legacy `micro-os-plus-iii-devices` repository isolated silicon device drivers (`soc/stm32f4xx`, `soc/rp2350`, `soc/bcm2837`) outside their corresponding CPU architecture repositories. This caused:

- Split architectural headers: Cortex-M33 register definitions lived in `cortexm`, but RP2350 SIO spinlocks lived in `devices`.
- Multi-repository synchronization locks: Modifying an interrupt controller required coordinated commits across two repositories before tests could build.
- Divergence from upstream xPack distribution packaging.

By folding device support directly into `cortexm`, `posix-arch`, `aarch32`, and `aarch64`, each architecture repository becomes a complete, self-contained hardware abstraction layer.

3.2 History-Preserving `git subtree` Extraction Protocol

Dissolution must **not** be performed via flat file copying; commit history must be 100% preserved. This is achieved using `git subtree add`:

```
cd "$WORK/micro-os-plus-iii-cortexm"

# Subtree add Cortex-M silicon devices from devices repo
git subtree add --prefix=devices "$WORK/micro-os-plus-iii-devices"
    master

# Restructure into standard architecture directories
mv devices/soc/stm32f4xx soc/stm32f4xx
mv devices/soc/rp2350 soc/rp2350
mv devices/drivers drivers
rm -rf devices

git add soc drivers
git commit -m "refactor(devices): fold Cortex-M SoC and drivers
    subtree into cortexm port"
```

The identical procedure is executed for `posix-arch` (folding `soc/native`), `aarch32` (folding `soc/bcm2837`), and `aarch64`.

3.3 CMake Target Realignment & Removal of `UOS_DEVICES_DIR`

Prior to dissolution, both `posix-arch` and `cortexm` expected an external sibling `micro-os-plus-iii-devices` directory and contained the following legacy boilerplate in their root `CMakeLists.txt`:

```
set (UOS_DEVICES_DIR "${uos_siblings}/micro-os-plus-iii-devices"
    CACHE PATH "μOS++ III devices working copy")

foreach (_dep_var IN ITEMS UOS_SMP_DIR UOS_DEVICES_DIR)
    ...
    if (NOT EXISTS "${${_dep_var}}/CMakeLists.txt")
        message (FATAL_ERROR "Cannot find the ${_repo} repository at ${${_dep_var}}...")
    endif ()
endforeach ()
```

```
add_subdirectory ("${UOS_DEVICES_DIR}" micro-os-plus-iii-devices)
```

If left unchanged after micro-os-plus-iii-devices is deleted, any subsequent CMake configure or test command fails immediately with:

```
CMake Error at /tmp/micro-os-plus-iii-posix-arch/CMakeLists.txt:
  Cannot find the devices repository at /tmp/micro-os-plus-iii-devices.
```

Realignment in micro-os-plus-iii-posix-arch/CMakeLists.txt

1. Remove UOS_DEVICES_DIR and its validation loop from the dependency checks.
2. Define the local block device and filesystem targets directly using the absorbed drivers/ and soc/native/ files:

```
# Dissolved devices support: hostfile block device and flatfs
if (NOT TARGET micro-os-plus::devices-hostfile)
  add_library (micro-os-plus-iii-devices-hostfile-interface INTERFACE)
  target_include_directories (
    micro-os-plus-iii-devices-hostfile-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/include"
      "${CMAKE_CURRENT_SOURCE_DIR}/soc/native/include"
  )
  target_compile_definitions (
    micro-os-plus-iii-devices-hostfile-interface INTERFACE
    SD_BACKEND_HOSTFILE
  )
  target_sources (
    micro-os-plus-iii-devices-hostfile-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/soc/native/src/sd_hostfile.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/flatfs.cpp"
  )
  add_library (micro-os-plus::devices-hostfile ALIAS
    micro-os-plus-iii-devices-hostfile-interface)
endif ()

# Compatibility alias for full devices suite (FatFs + block device)
if (NOT TARGET micro-os-plus::devices)
  add_library (micro-os-plus-iii-devices-interface INTERFACE)
  target_include_directories (
    micro-os-plus-iii-devices-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/include"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs"
  )
  target_sources (
    micro-os-plus-iii-devices-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/sd.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/flatfs.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/usb_dwc2.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/ff.c"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/diskio_sd.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/fatfs_hw.cpp"
  )
  add_library (micro-os-plus::devices ALIAS micro-os-plus-iii-devices-interface)
endif ()
```

Realignment in micro-os-plus-iii-cortexm/CMakeLists.txt

1. Remove U0S_DEVICES_DIR and its validation loop from the dependency checks.
2. Define the local STM32F4xx and RP2350 silicon support targets directly using the absorbed soc/ files:

```
# Dissolved SoC targets: STM32F4xx and RP2350
if (NOT TARGET micro-os-plus::soc-stm32f411xe)
    add_library (micro-os-plus-iii-soc-stm32f4xx-common-interface
        INTERFACE)
    target_include_directories (
        micro-os-plus-iii-soc-stm32f4xx-common-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/include"
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/include/cmsis"
    )
    target_sources (
        micro-os-plus-iii-soc-stm32f4xx-common-interface
        INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/src/
            system_stm32f4xx.c"
    )

    add_library (micro-os-plus-iii-soc-stm32f411xe-interface INTERFACE)
    target_link_libraries (
        micro-os-plus-iii-soc-stm32f411xe-interface
        INTERFACE micro-os-plus-iii-soc-stm32f4xx-common-interface
    )
    target_sources (
        micro-os-plus-iii-soc-stm32f411xe-interface
        INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/src/
            vectors_stm32f411xe.c"
    )

    add_library (micro-os-plus::soc-stm32f411xe
        ALIAS micro-os-plus-iii-soc-stm32f411xe-interface)

    add_library (micro-os-plus-iii-soc-stm32f412rx-interface INTERFACE)
    target_link_libraries (
        micro-os-plus-iii-soc-stm32f412rx-interface
        INTERFACE micro-os-plus-iii-soc-stm32f4xx-common-interface
    )
    target_sources (
        micro-os-plus-iii-soc-stm32f412rx-interface
        INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/stm32f4xx/src/
            vectors_stm32f412rx.c"
    )

    add_library (micro-os-plus::soc-stm32f412rx
        ALIAS micro-os-plus-iii-soc-stm32f412rx-interface)
endif ()

if (NOT TARGET micro-os-plus::soc-rp2350)
    add_library (micro-os-plus-iii-soc-rp2350-interface INTERFACE)
    target_include_directories (
        micro-os-plus-iii-soc-rp2350-interface
        INTERFACE
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/rp2350/include"
            "${CMAKE_CURRENT_SOURCE_DIR}/soc/rp2350/include/cmsis"
    )
    target_sources (
        micro-os-plus-iii-soc-rp2350-interface
        INTERFACE "${CMAKE_CURRENT_SOURCE_DIR}/soc/rp2350/src/
            system_rp2350.c"
    )

    add_library (micro-os-plus::soc-rp2350
        ALIAS micro-os-plus-iii-soc-rp2350-interface)
endif ()
```

Realignment in aarch32 and aarch64

Both `micro-os-plus-iii-aarch32/CMakeLists.txt` and `micro-os-plus-iii-aarch64/CMakeLists.txt` compile the dissolved `soc/bcm2837` sources locally into `micro-os-plus::aarch32` and `micro-os-plus::aarch64` interface targets, with zero external dependency on devices.

Furthermore, following the dissolution of `micro-os-plus-iii-devices`, tests exercising the SD host (`sd_test`, `smp-mat-sdcard-test`), USB controller (`usb_test`), and file systems require the `micro-os-plus::devices` target. Both AArch32 and AArch64 declare this target locally from their absorbed `drivers/` and `soc/` trees:

```
# Dissolved devices support: SD host, USB, and FatFs stack
if (NOT TARGET micro-os-plus::devices)
  add_library (micro-os-plus-iii-devices-interface INTERFACE)
  target_include_directories (
    micro-os-plus-iii-devices-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/include"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs"
  )
  target_sources (
    micro-os-plus-iii-devices-interface
    INTERFACE
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/sd.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/flatfs.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/src/usb_dwc2.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/ff.c"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/diskio_sd.cpp"
      "${CMAKE_CURRENT_SOURCE_DIR}/drivers/fatfs/fatfs_hw.cpp"
  )
  add_library (micro-os-plus::devices ALIAS micro-os-plus-iii-devices-interface)
endif ()
```

Duplicate Port Target Prevention in Platform Harness

In the `μOS++` test framework (`tests/cmake/tests-main.cmake`), the architecture port under test is added directly via `add_subdirectory` ("`$UOS_<PORT>_DIR`" "`port-bin`"). Therefore:

- The architecture port (`micro-os-plus-iii-aarch32` or `micro-os-plus-iii-aarch64`) must **never** be listed in `dependencies-folders.cmake` (`xpack_dependencies_folders`).
- Listing the port in `dependencies-folders.cmake` causes CMake policy CMP0002 violations (`add_library` cannot create target ... because another target with the same name already exists).
- Platforms list only test suite directories and portable xPacks (`mutex-stress`, `rtos-apis`, `arm-cmsis-rtos-validator`, `chan-fatfs`) in `dependencies-folders.cmake`.

Shared Test Support Scaffolding Realignment (`test_smp1` → `tests/smp-support`)

When the shared test scaffolding was consolidated into `tests/smp-support/` (`hw_result.hpp` and `board-contract.cpp`):

- `micro-os-plus::test-support` in `micro-os-plus-iii/CMakeLists.txt` exposes `${CMAKE_CURRENT_SOURCE_DIR}/tests/smp-support/include` (with a fallback to `test_smp1/include`).
- A backwards-compatibility symlink `test_smp1 -> tests/smp-support` is maintained in the root of `micro-os-plus-iii` so that legacy test runners (`run-hw.sh`, `run-host.sh`, `run-qemu.sh`) and scripts referencing `$SMP_DIR/test_smp1/` execute without error.

3.4 Phase 0 Verification Gate

```
# Verify all existing uniprocessor boards configure and build cleanly
cd "$WORK/micro-os-plus-iii"
xpm run test-all # Must remain 72/72 PASS
```

3.5 Permanent Deletion of the Legacy devices Repository and Working Directory

Once the subtree extraction into cortexm, posix-arch, aarch32, and aarch64 is complete and verified, the micro-os-plus-iii-devices repository and working directory have fulfilled their transitional purpose.

The micro-os-plus-iii-devices Git clone/directory must be **permanently deleted from the active workspace**:

```
# Permanently delete the dissolved devices repository clone from the
workspace
rm -rf "$WORK/micro-os-plus-iii-devices"
```

Key Architectural Rules Post-Deletion:

1. **Zero External Device Dependency:** No architecture repository or kernel build configuration may reference or depend on an external micro-os-plus-iii-devices directory.
 2. **Self-Contained Ports:** Silicon drivers (soc/*) and neutral drivers (drivers/*) are compiled and maintained directly inside each architecture repository (cortexm, posix-arch, aarch32, aarch64).
 3. **Retired Repository:** The standalone micro-os-plus-iii-devices repository is officially retired and archived; it is never propagated into future xpack-development or smp deployment cycles.
-

4. Phase 1: Single-Core Defect Corrections & Hardening (PR #1 through PR #22)

Phase 1 isolates all bugfixes, standard conformance corrections, and allocator hardening that are **pure uniprocessor improvements**. Landing these 22 commits first guarantees that single-core $\mu\text{OS}++$ is completely defect-free before any multi-core scheduling logic is introduced.

Summary Table of Phase 1 Pull Requests

PR #	Subject / Theme	File(s) Touched	Problem & Failure Mode Averted	Verification Gate
#1	POSIX dirent.h ISO C Conformance	include/cmsis-plus/posix/dirent.h	Empty struct typedef struct { ; } DIR; violates ISO C99/C11 §6.7.2.1; triggers -Wextra-semi.	72/72 pass; clean -Wextra-semi
#2	List Iterator Access Semantics	include/cmsis-plus/utls/lists.h	node_>next accessed as data member instead of method node_>next(), breaking C++17/20 concepts.	72/72 pass; template concepts clean
#3	Newlib Syscall Return Types	posix-io/c-syscalls-aliases-standard.h	Newlib 64-bit uses _READ_WRITE_RETURN_TYPE (int or _ssize_t), causing conflicting weak alias signatures.	72/72 pass; reentrant tests clean
#4	Export Thread Suspend Symbol	src/rtos/os-thread.cpp	Out-of-line inline void suspend() prevented symbol emission; undefined reference in C API wrapper.	72/72 pass; wrapper links cleanly
#5	Memory Allocator Usable Size	os-memory.*, malloc.cpp, first-fit-top.*	Missing do_usable_size() broke realloc(); calloc() lacked multiplication overflow checks.	72/72 pass; usable size verified
#6	C++17 Aligned Allocation	src/libcpp/new.cpp	Missing standard operator new/delete(align_val_t) overloads for over-aligned types.	72/72 pass; aligned new clean
#7	Static Error Category Storage	src/libcpp/system-error.cpp	Temporary category objects left dangling pointers inside throw std::system_error.	72/72 pass; static lifetime clean
#8	Chrono 64-Bit Duration Overflow	src/libcpp/chrono.cpp	cycles * 1e9 / freq overflowed 64-bit int after ~13 minutes. Decompose into seconds and remainder.	72/72 pass; chrono math overflow-free
#9	CMSIS-RTOS C Wrapper Cleanups	src/rtos/os-c-wrapper.cpp	Default timer must be one-shot; polymorphic mutex delete; cast millisec to uint64_t prior to * 1000.	72/72 pass; validator 60/60 clean
#10	POSIX I/O FD Table Mutexing	src/posix-io/file-descriptors-manager.cpp	Unprotected descriptor table corrupted under concurrent multi-threaded open()/close().	72/72 pass; concurrent I/O stress pass
#11	POSIX I/O Deferred List Locking	posix-io/file-system.h, net-stack.h	Concurrent recycling of deferred file/socket nodes caused corrupted intrusive list links.	72/72 pass; list recycling clean
#12	Block Device Sector Size Check	src/posix-io/block-device.cpp	Inverted check in vfcntl: tested != 0 instead of == 0,	72/72 pass; valid block

PR #	Subject / Theme	File(s) Touched	Problem & Failure Mode Averted	Verification Gate
			falsely rejecting valid block devices.	size returned
#13	ARMv8-M Semihosting Traps	include/cmsis-plus/arm/semihosting.h	Missing __ARM_ARCH_8M_MAIN__ macro guards and SEMIHOST_TRAP_HLT support.	72/72 pass; ARM semihosting pass
#14	ARMv8-M Exception Handlers	src/startup/exception-handlers.c	SCB register setup and fault dump routines lacked __ARM_ARCH_8M_MAIN__ guards.	72/72 pass; fault handlers compile
#15	Semihosting fstat() & Mirror	src/semihosting/c-syscalls-semihosting.cpp	Assign S_IFCHR only when mode unset; add weak os_board_console_mirror() hook.	72/72 pass; console mirror clean
#16	Timer Callback Decoupling	internal/os-lists.cpp, os-timer.cpp	Executing timer callbacks inside critical section caused deadlocks; drift catch-up guard.	72/72 pass; timer callback mutexes clean
#17	Mutex Priority Ceiling Protocol	src/rtos/os-mutex.cpp	Enforce ceiling before acquiring ownership; track maximum boost among multiple waiters.	72/72 pass; mutex-stress-test pass
#18	Thread Lifecycle State & Reaper	os-c-decls.h, os-thread.h, os-idle.cpp	Add state::destroying = 7; idle thread reaper safely validates no core is running thread before free.	72/72 pass; thread reaper green
#19	CondVar Lost-Signal Elimination	src/rtos/os-condvar.cpp, os-condvar.h	Atomic link node & mutex unlock under scheduler::critical_section; bind condvar to clock.	72/72 pass; condvar concurrency green
#20	std::thread Handle Sync	thread.cpp.h, thread_internal.h	thread::join() waits on native thread handle; retain functor object to eliminate use-after-free.	72/72 pass; functor lifetime verified
#21	MQueue Preemptive Reschedule	src/rtos/os-mqueue.cpp	Call port::scheduler::reschedule() immediately after send/receive wakeups.	72/72 pass; queue preemption green
#22	High-Res Clock Port Sync	os-decls.h, os-clocks.cpp, Port inlines	Declares has_hardware_counter(); fast-paths now(); SysTick ICSR pending overflow inspection.	72/72 pass; ICSR pending fix verified

5. Phase 2: Common Multi-Core SMP Kernel Infrastructure & Host Runtime (PR #23 / Step 14)

5.1 Architectural Foundations of the SMP Kernel

Phase 2 constitutes the architectural core of the entire SMP migration. It transitions the kernel from a single-CPU RTOS to a Symmetric Multiprocessing engine capable of executing threads concurrently across NN physical cores.

Key data structure transformations in `include/cmsis-plus/rtos/os-sched.h` and `src/rtos/os-core.cpp`:

- **Per-CPU Current Threads:** Replaces scalar `current_thread_` with an array indexed by `port_cpu_id()`:

```
#if defined(OS_USE_SMP_SCHEDULER)
extern thread* volatile current_thread_[OS_NCPU];
#else
extern thread* volatile current_thread_;
#endif
```

- **Per-CPU Ready Lists:** Replaces the monolithic ready list with `ready_list_[OS_NCPU]`, eliminating global queue lock bottlenecking during normal thread execution.
- **Recursive Kernel Spinlock (`_smp_klock`):** Protects scheduler state transitions. Backed by architecture hardware spinlocks (e.g. RP2350 SIO spinlock 0, Cortex-M33 LDREX/STREX, or POSIX mutex).

5.2 The 5-Stage Stack Pointer Claim/Publish Protocol

To prevent fatal stack corruption when two cores attempt to switch to or from the same thread simultaneously, the scheduler implements a strict 5-stage handshake:

1. **Claim:** The incoming thread stack pointer is atomically set to `nullptr` on the target core, marking it actively running.
2. **Spill:** The outgoing thread's callee-saved registers are pushed to its current stack.
3. **Switch:** Hardware Stack Pointer register (SP) is updated to the incoming thread's stack.
4. **Publish:** The outgoing thread's saved stack pointer is published to `from->context_.port_.stack_ptr = sp` **only after** state is completely saved.
5. **Restore:** Callee-saved registers are popped from the new stack, and thread execution resumes.

5.3 Prohibiting Context Switch While Holding Spinlock

A primary invariant in `μOS++` SMP: **A hardware context switch must never be executed while holding `_smp_klock`.** Context switching while holding the kernel spinlock causes deadlocks across cores. Instead, `internal_switch_threads()` schedules the switch, marks the CPU as requiring rescheduling, releases `_smp_klock`, and allows the architecture PendSV interrupt or signal handler to execute the actual register swap.

5.4 Phase 2 Verification Gates

1. **The undef Single-Core Invariant:** Running undef -UOS_USE_SMP_SCHEDULER on the Phase 2 kernel source must produce a tree that is **100% byte-identical to Step 22**. Uniprocessor builds incur zero code bloat and zero overhead.
2. **72/72 Uniprocessor Test Gate:** xpm run test-all passes with 100% green status.
3. **Genuine Dual-Core Host Verification (native-smp):** Executes tests/platforms/native-smp/ under Linux host using 2 OS threads as virtual CPU cores:

```
cd "$WORK/micro-os-plus-iii/tests"
xpm run test-native-smp
# Must execute and pass:
# 1. rtos-apis-test (dual-core concurrency)
# 2. mutex-stress-test (contention across cores)
# 3. cmsis-os-validator-test (60/60 test cases)
```

6. Phase 3: Semantic Versioning, Port Releases & Silicon Extensions (PR #24 through PR #27)

6.1 PR #24: Port Package Semantic Releases

Publishes the formal multi-core hardware abstraction contract:

- `micro-os-plus-iii-posix-arch`: Version bumped from 1.0.1 → 1.1.0.
- `micro-os-plus-iii-cortexm`: Version bumped from 1.1.0 → 1.2.0. Local semantic git tags (v1.1.0, v1.2.0) are generated for repeatable downstream dependency resolution.

6.2 PR #25: Execution of Devices Subtree Dissolution, CMake Realignment & Repository Deletion

Finalizes the historical merge of `soc/*` and `drivers/*` into the owning architecture ports, establishing clean build targets and permanent ALIAS compatibility shims.

Key Implementation Actions:

1. **Subtree Merge:** Folds `soc/stm32f4xx` and `soc/rp2350` into `cortexm`, `soc/native` and `drivers/*` into `posix-arch`, and `soc/bcm2837` into `aarch32/aarch64`.
2. **CMakeLists.txt Realignment:**
 - In `posix-arch/CMakeLists.txt`: Remove `U0S_DEVICES_DIR` dependency check. Declare `micro-os-plus::devices-hostfile` (from `soc/native/src/sd_hostfile.cpp` and `drivers/src/flatfs.cpp` with `SD_BACKEND_HOSTFILE`) and `micro-os-plus::devices` directly.
 - In `cortexm/CMakeLists.txt`: Remove `U0S_DEVICES_DIR` dependency check. Declare `micro-os-plus::soc-stm32f411xe`, `micro-os-plus::soc-stm32f412rx`, and `micro-os-plus::soc-rp2350` directly from the local `soc/` paths.
3. **Repository Deletion:** Immediately upon completion of the subtree graft and CMake realignment, the `micro-os-plus-iii-devices` repository folder and Git repository are **permanently deleted from the workspace**, retiring the standalone repository and guaranteeing that all downstream projects build and test 100% self-contained.

6.3 PR #26: Modern Multi-Core Silicon Ports (Cortex-M33 & RP2350)

Introduces genuine silicon hardware multi-core implementations:

- **ARM Cortex-M33 Port:** Supports ARMv8-M architecture, Message Handling Unit (MHU) Inter-Processor Interrupts (IPI), and per-core SysTick clocks.
- **Raspberry Pi RP2350 Port:** Implements recursive `_smp_klock` backed by SIO Hardware Spinlock 0 (*(volatile uint32_t*) (0xD0000100)), Core 1 wake-up mailbox, and FIFO IRQ 25.
- **Kernel IPI Dispatching:** In `src/rtos/os-thread.cpp`, dispatches wake-up IPIs whenever a thread unblocks on an alternate core (`OS_INTEGER_RTOS_PORT_NCPU > 1`).

6.4 PR #27: Modular Add-Only CMake Build Architecture

Splits the monolithic CMake interface into clean, granular sub-targets:

- `micro-os-plus::iii-core`: Bare kernel RTOS engine without POSIX I/O.
 - `micro-os-plus::iii-posix-io`: File system, mount tables, and character devices.
 - `micro-os-plus::port-smp-decls`: Pure SMP forward declarations for thin ports.
 - Adds cross-compilation toolchain profiles (`cmake/toolchains/arm-none-eabi.cmake`, `cmake/toolchains/aarch64-none-elf.cmake`).
 - Adds QEMU 2-core platform tests/`platforms/2xcortex-m33/` (`mps2-an521 --cpu cortex-m33 --smp 2`).
-

7. Phase 4: Multi-Core Test Suites, Multi-Arch Platforms & CI Harness (PR #28 & PR #29)

7.1 PR #28: Multi-Core FPU Stress Suite & Scaffolding Relocation

- **FPU Context Switch Stress Suite** (`tests/sources/fp-switch/`): 6 threads computing mathematical floating-point series across cores under aggressive 1ms preemptive slicing (3000 ticks). Detects missing hardware FPU lazy-stacking or register corruptions.
- **Scaffolding Relocation**: Moves legacy root `test_smpl/` into `tests/smp-support/`, cleanly exported as `micro-os-plus::test-support` providing standardized test reporting and hardware exit registers.

7.2 PR #29: Multi-Architecture 4-Core Platforms & Composite CI Action

- **AArch32 & AArch64 4-Core Platforms**: Implements QEMU raspi3b platforms (`tests/platforms/aarch32-rpi3b/`, `tests/platforms/aarch64-rpi3b/`) executing across 4 ARM Cortex-A53 cores.
- **Composite Test Action** (`test-smp-all`): Registered in `tests/package.json`, executing the complete multi-core test matrix in one command:

```
"test-smp-all": {  
  "description": "Run all SMP test suites across Native, Cortex-M33, and AArch32/64",  
  "actions": [  
    "xpm run test-native-smp",  
    "xpm run test-2xcortex-m33-cmake",  
    "xpm run test-aarch32-rpi3b",  
    "xpm run test-aarch64-rpi3b"  
  ]  
}
```

8. Phase 5: Documentation Suite, Master Reconciliation & Silicon Wiring (PR #30 through PR #32)

8.1 PR #30: Comprehensive Documentation & Specification Runbooks Suite

Integrates all architectural runbooks and migration specifications into docs/render-pdfs.sh. Builds the complete 16-document PDF publication suite.

8.2 PR #31: Final Master Reconciliation & Repository Cleansing

- Merges upstream origin/xpack-development cleanly.
- Restores .github/workflows/ci.yml, README.md, LICENSE, and Doxygen templates to be **100% byte-identical** to upstream baseline.
- Purges temporary transition scripts (scripts/smp/).

8.3 PR #32: Dedicated Raspberry Pi Pico 2 Physical Hardware Platform Wiring

Adds standalone hardware platform tests/platforms/cortexm-pico2/ enabling direct OpenOCD SWD flashing and UART verification on physical RP2350 silicon. Implements Core 1 secondary boot handshake via SIO mailbox registers (sio_hw->fifo_wr / sio_hw->fifo_rd).

9. Operator's Execution Manual: The Step-by-Step Lifting Loop

Superseded. The script-driven loop below (new-step.sh, chunks/stepNN.sh, verify-step.sh, advance-step.sh) is kept only as a historical record. The authoritative procedure is the commit-only cycle in §12. Experience with this loop showed that the recipes mix subjects: a git checkout golden/smp -- <files> lift staged the files of PR #2 and PR #3 into the commit for PR #1. The scripts also leave transition tooling in the repository, which then has to be removed again (PR #31).

9.1 The Golden 4-Command Development Cycle per Step

For each of the 32 pull requests, the developer executes a standardized four-command loop:

```
+-----+ +-----+
+-----+ +-----+
| 1. NEW STEP | ---> | 2. APPLY CHUNK | ---> | 3. VERIFY
GATE | ---> | 4. COMMIT / ADV |
| Fork step branch | | Lift clean diff | | Compile &
Test | | Atomic git commit|
+-----+ +-----+
+-----+ +-----+
```

Step 1: Fork the Step Branch

```
cd "$WORK/micro-os-plus-iii"
# Creates step/NN branched from step/NN-1 (or xpack-development at
# Step 1)
bash scripts/smp/new-step.sh <NN>
```

Step 2: Apply the Step's Isolated Chunk

```
# Sourced directly from golden reference repo in ~/Work2/micro-os-
plus/
bash scripts/smp/chunks/step<NN>.sh
# Or inspect unified diff:
bash scripts/smp/show-chunk.sh <NN> <file...>
```

Step 3: Execute the Verification Gate

```
# Validates pristine rules, unifdef projection, and runs 72/72 test
matrix
bash scripts/smp/verify-step.sh <NN>
```

Step 4: Commit and Advance

```
# Creates atomic git commit with conventional commit header
bash scripts/smp/advance-step.sh <NN> "commit message description"
```

9.2 Stacking Pull Requests on GitHub

When submitting the 32 steps to GitHub:

1. Push each local step branch: git push -u origin step/<NN>.

2. Open Pull Request on GitHub:
 - For PR #1: Target base xpack-development ←\leftarrow Head step/01.
 - For PR #2: Target base step/01 ←\leftarrow Head step/02.
 - For PR #N: Target base step/<N-1> ←\leftarrow Head step/<N>.
3. When upstream merges PR #1 into xpack-development, use GitHub's web UI to retarget PR #2's base from step/01 to xpack-development.

9.3 Cherry-Picking Commits Back into xpack-development

Because each commit is strictly self-contained and touches only files relevant to its single theme, any individual commit or group of commits can be cherry-picked directly into xpack-development:

```
cd "$WORK/micro-os-plus-iii"
git checkout xpack-development
git fetch origin

# Example: Cherry-pick PR #1 (dirent) and PR #2 (lists) directly
git cherry-pick <PR_1_COMMIT_HASH>
git cherry-pick <PR_2_COMMIT_HASH>

# Verify baseline tests remain 100% green
xpm run test-all
```

10. Comprehensive Verification Matrix & Acceptance Gates

Every pull request satisfies an explicit, automated acceptance gate prior to being advanced:

PR ID	Theme & Domain	Repositories	Self-Contained?	Automated Gate Command	Acceptance Pass Criteria
PR #1	POSIX dirent.h Conformance	Kernel	standalone	xpm run test-all	72/72 tests pass; -Wextra-semi clean
PR #2	List Iterator Member Semantics	Kernel	standalone	xpm run test-all	72/72 tests pass; template concepts clean
PR #3	Newlib Syscall Return Types	Kernel	standalone	xpm run test-all	72/72 tests pass; newlib reentrant tests pass
PR #4	Thread Suspend Symbol Export	Kernel	standalone	xpm run test-all	72/72 tests pass; C wrapper resolves symbol
PR #5	Memory Allocator Usable Size	Kernel	standalone	xpm run test-all	72/72 tests pass; malloc_usable_size verified
PR #6	C++17 Aligned Allocation	Kernel	standalone	xpm run test-all	72/72 tests pass; aligned operator new clean
PR #7	Static Error Category Storage	Kernel	standalone	xpm run test-all	72/72 tests pass; static storage eliminates dangling ptrs
PR #8	Steady Clock Duration Overflow	Kernel	standalone	xpm run test-all	72/72 tests pass; chrono math overflow-free
PR #9	CMSIS-RTOS C Wrapper Cleanups	Kernel	standalone	xpm run test-all	72/72 tests pass; validator (60/60) clean
PR #10	POSIX I/O FD Table Mutex	Kernel	standalone	xpm run test-all	72/72 tests pass; concurrent I/O stress green
PR #11	POSIX I/O Deferred List Locking	Kernel	standalone	xpm run test-all	72/72 tests pass; list recycling thread-safe
PR #12	Block Device Sector Size Check	Kernel	standalone	xpm run test-all	72/72 tests pass; valid block size returned
PR #13	ARMv8-M Semihosting Traps	Kernel	standalone	xpm run test-all	72/72 tests pass; ARM semihosting tests pass
PR #14	ARMv8-M Exception Handlers	Kernel	standalone	xpm run test-all	72/72 tests pass; ARMv8-M fault vectors compile
PR #15	Semihosting fstat() & Mirror	Kernel	standalone	xpm run test-all	72/72 tests pass; mode preserved & console mirror hook
PR #16	Software Timer Decoupling	Kernel	standalone	xpm run test-all	72/72 tests pass; timer callback mutexes clean
PR #17	Mutex Priority Ceiling	Kernel	standalone	xpm run test-all	

PR ID	Theme & Domain	Repositories	Self-Contained?	Automated Gate Command	Acceptance Pass Criteria
					72/72 tests pass; mutex-stress-test passes
PR #18	Thread Lifecycle & Reaper	Kernel	standalone	xpm run test-all	72/72 tests pass; dynamic thread reaper green
PR #19	CondVar Lost-Signal Elimination	Kernel	standalone	xpm run test-all	72/72 tests pass; condvar concurrency stress green
PR #20	std::thread Handle Sync	Kernel	standalone	xpm run test-all	72/72 tests pass; functor closure lifetime verified
PR #21	MQueue Preemptive Reschedule	Kernel	standalone	xpm run test-all	72/72 tests pass; message queue preemption green
PR #22	High-Res Clock Port Sync	K + C + P	atomic	xpm run test-all	72/72 tests pass; ICSR pending fix verified
PR #23	Multi-Core SMP Core & Host	K + C + P	atomic	scripts/smp/verify-step.sh 14	unifdef zero-diff + 72/72 + native-smp (3/3) PASS
PR #24	Port Semantic Releases	P + C	standalone	git tag -l	v1.1.0 and v1.2.0 local tags generated
PR #25	Devices Subtree Dissolution	P + C + Arch	standalone	xpm run build ...	Subtree merged; CMake ALIAS targets resolve; devices folder deleted from workspace
PR #26	Modern Silicon Ports	C + K	add-only	scripts/smp/verify-step.sh 25	M33/RP2350 port targets compile cleanly
PR #27	Modular Add-Only CMake	K + C	add-only	scripts/smp/verify-step.sh 26	72/72 baseline + 2xcortex-m33 (4/4) PASS
PR #28	FPU Context Switch Stress	Kernel	add-only	scripts/smp/verify-step.sh 27	fp-switch passes on 2xcortex-m33 (4/4)
PR #29	Multi-Arch 4-Core Platforms	K + A32 + A64	add-only	scripts/smp/verify-step.sh 28	test-smp-all passes (Native, M33, AArch)
PR #30	Documentation Suite	Kernel	standalone	docs/render-pdfs.sh	All PDFs compile cleanly without warnings
PR #31	Final Merge & Cleansing	All Repos	clean-merge	bash scripts/smp/finalize.sh	Upstream CI metadata 100% byte-identical
PR #32	Pico 2 Hardware Platform	K + C	add-only	xpm run build --config cortexm-pico2	Pico 2 platform compiles for hardware SWD

10.2 Final Parity Verification with Golden Source

At the conclusion of Step 32:

- The rebuilt smp branch in \$WORK/micro-os-plus-iii is compared against /home/dan/Work2/micro-os-plus/micro-os-plus-iii:

```
diff -ruN -x ".git" "$WORK/micro-os-plus-iii" "/home/dan/Work2/  
micro-os-plus/micro-os-plus-iii"
```

- **Acceptance Invariant:** The source code trees across all five repositories (micro-os-plus-iii, cortexm, posix-arch, aarch32, aarch64) achieve **100% functional and structural parity** with the reference implementation in ~/Work2/micro-os-plus/, but possess a **clean, granular, 32-commit GitHub Pull Request history** ready for immediate upstream integration.
-

Part II — Commit-Only Migration Procedure (Authoritative)

11. Principles: Only Commits, One Subject per Commit, No Scripts

11.1 Why Part I's script loop is replaced

Part I (§9) drives the rebuild with `scripts/smp/*.sh` and `chunks/stepNN.sh` recipes. A dry run on clean clones of `github.com/dan-maio/*` showed three structural problems with it:

1. **Recipes mix subjects.** A recipe lifts whole files with `git checkout golden/smp -- <files>`. When that happens, the lift *stages* every lifted file. Step 01's recipe staged `lists.h` (PR #2) and `c-syscalls-aliases-standard.h` (PR #3) into the commit for PR #1, so the commit no longer matched its PR.
2. **The tooling becomes part of the history.** `scripts/smp/`, its `chunks` and their `__pycache__` are committed, and later have to be removed again (PR #31). A GitHub reviewer sees tooling churn that has nothing to do with the kernel.
3. **A step's result is not reviewable before it exists.** What lands is whatever the script produced at run time. A reviewer cannot read a recipe and know the exact resulting diff.

Decision: the rebuild uses **git commits only**. Every commit is written as the pull request it will become: one subject, a self-contained message, and a diff that is reviewable on its own. The procedure needs no shell or Python script. The only commands are `git` (including the interactive hunk selection built into `git`) and `xpm` (the μ OS++ test framework).

11.2 The ten rules

#	Rule	Consequence
R1	One subject per commit: one correction, one feature, one test, one board, one platform, or one document.	A file touched by several subjects is lifted hunk by hunk (<code>git checkout -p</code> , §12.2), never whole.
R2	A commit only modifies or adds files. It never deletes or renames an upstream file.	Golden's deletions (<code>.github/</code> , <code>doxygen/</code> , <code>inspiration/</code> , <code>templates/</code> , <code>.vscode/</code> , <code>.settings/</code> , <code>CHANGELOG.md</code> , <code>README*.md</code> , <code>LICENSE</code> → <code>LICENSE-micro-os-plus-iii</code> , <code>package-lock.json</code> , <code>CLAUDE.md</code> , <code>arch.npmignore/LICENSE/README.md</code> , <code>posix-arch test/trace-*</code>) are not reproduced . PR #31's "restore to upstream" therefore has nothing to do.
R3	No procedure tooling is committed.	<code>scripts/smp/**</code> and <code>tests/xpacks/**</code> (installed packages, created by <code>xpm install</code>) are never committed.
R4	Product files may be shell, procedure files may not.	Test runners that <code>ctest</code> executes (<code>test_smpl/run-{host,qemu,hw}.sh</code> , <code>test/boards/*/hw.sh</code> , <code>test/qemu.sh</code> , <code>test/boards/native/run.sh</code>) are product files. They are committed with the test infrastructure that calls them.
R5	A fix to code that this series introduces is folded into the commit	No PR ships a known bug that a later PR repairs. Fixes to <i>upstream</i> code are separate commits (the Phase 1 corrections).

#	Rule	Consequence
	that introduces that code.	
R6	The provider lands before the consumer.	A port symbol lands before the kernel code that calls it, and kernel declarations land before port code that implements them. When neither order compiles alone, the commits form a group . The gate runs at the group's last commit, and every message in the group names the group (Group: G-hrclock).
R7	Every commit compiles in the single-core configurations.	New SMP code sits under <code>#if defined(OS_USE_SMP_SCHEDULER).unifdef -UOS_USE_SMP_SCHEDULER</code> of any SMP commit gives the tree of its parent.
R8	New build files grow with their sources.	A commit that adds a .cpp also adds its line to the port's CMakeLists.txt. No commit lists a file that does not exist yet.
R9	Tests are added one per commit.	The port test builders (test/CMakeLists.txt) and the kernel platforms register <i>every directory present</i> under test/<board>/. One new test directory therefore equals one commit and one new ctest entry.
R10	Every gate is an xpm action of the µOS+ + test framework.	§13.3. No ad-hoc gate script exists.

11.3 Corrections to Part I, checked against the golden smp branches

These facts were read from ~/Work2/micro-os-plus/* (xpack-development..smp, read-only):

Part I says	Golden smp actually has	Part II rule
Clone from github.com/micro-os-plus/* (§2.2)	Fork github.com/dan-maio/*. On aarch64, local xpack-development is 366c957 but GitHub origin/xpack-development is fe904c0.	Clone from dan-maio. Before starting, compare the aarch64 base with <code>git log --oneline 366c957...fe904c0</code> and record which one was used.
xpm link @micro-os-plus/cortexm (§2.4)	The package names are @micro-os-plus/micro-os-plus-iii-{posix-arch,cortexm,aarch32,aarch64}.	Link them per configuration through <code>xpm run install --config <cfg></code> ; each *-actions configuration already runs xpm link. Do not use link-deps-all: it aborts on @micro-os-plus/build-helper.
tests/platforms/native-smp/, test-native-smp (§5.4)	No such platform. tests/platforms/native builds and runs posix-arch's own test/native/* (rtos-apis, mutex-stress, flatfs, mutex-ceiling, smp-*) through test_smpl/run-host.sh.	Native SMP tests are posix-arch test directories (§16.3). They run under <code>xpm run test-native-cmake</code> .
test-smp-all action (§7.2)	The action is test-smp-cmake, and it is part of test-all (native + cortex + smp).	The first SMP platform commit creates test-smp-cmake and adds it to test-all. Each later platform commit appends one line to it (§15.6).
test_smpl/ → tests/smp-		

Part I says	Golden smp actually has	Part II rule
support/ plus a symlink (§3.3, §7.1)	test_smp/ is a real directory at the kernel root, and the platforms reference ../test_smp.	test_smp/ is added as is (K-H5*). No relocation, no symlink.
Release bumps posix-arch 1.1.0, cortexm 1.2.0 (§6.1)	package.json versions are unchanged: posix-arch 1.0.1, cortexm 1.1.0, aarch32/64 0.1.0.	A version bump is an optional last commit per port (chore(release)), made only when publishing.
PR #25 = devices dissolution (§6.2)	This is the same work as Phase 0.	Done once, as stage S0 (§14).
PR #23 must be one atomic K+C+P commit (§5, files-modif-by-step)	The kernel SMP code is guarded (R7), so single-core builds do not see it.	It is split into themed commits per repository (§15.3, §16.2, §17.2). SMP builds are expected to be green only at the end of stage S2, which is the S2 gate.
Final parity is "100% identical" (§10.2)	Identity is impossible by design: R2/R3 keep upstream files that golden deletes, and the arch repos gain soc/, drivers/ that golden only references.	The parity check lists the intended differences (§19.2). Every other difference is a defect.
devices subtree is added under devices/ and then moved (§3.2)	The devices repository (github/smp = e8e39d5) has include/, src/, fatfs/ and soc/<chip>/ at its top level.	It is imported with plain add commits into drivers/include, drivers/src, drivers/fatfs, soc/<chip> (§14).

12. The Commit Cycle (Manual, No Scripts)

12.1 Workspace

```

export WORK=/tmp                                # or any empty directory
cd "$WORK"
for r in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-
    iii-posix-arch \
    micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64 micro-os-
    plus-iii-devices; do
    git clone https://github.com/dan-maio/$r.git
done
for r in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-
    iii-posix-arch \
    micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64; do
    git -C "$WORK/$r" remote add golden "$HOME/Work2/micro-os-plus/
    $r" # read-only reference
    git -C "$WORK/$r" fetch golden
    smp                                     # fetch runs in $WORK,
    not in golden
    git -C "$WORK/$r" switch -c smp origin/xbpack-development
done

```

The golden repositories are **only read**. Every write (fetch, switch, commit) runs in \$WORK.

12.2 Building one commit: four git-native ways to lift content

Need	Command	Note
New file, whole		

Need	Command	Note
	<code>git checkout golden/smp -- <path></code>	Then <code>git restore --staged <path></code> if other paths are staged. Commit with an explicit path list (step 3 below).
Part of a modified file	<code>git checkout -p golden/smp -- <path></code>	Interactive: answer y only for the hunks of this commit's subject.
Hunk needs editing (two subjects in one hunk)	<code>git checkout -p then e</code>	Or edit in the editor. The diff to golden for the remaining subject stays visible with <code>git diff golden/smp -- <path></code> .
Content from the devices repository	<code>git -C ../micro-os-plus-iii-devices show e8e39d5:<path> > <new path></code>	Or <code>git read-tree --prefix=soc/rp2350/ -u <tree-ish></code> after fetching devices as a remote (§14.2).

12.3 The per-commit cycle

1. **BRANCH** `git switch -c pr/<ID>-<slug>` (stacked on the previous pr/ branch)
2. **LIFT** one subject only (§12.2)
3. **COMMIT** `git add <exact paths> && git commit` (message template §12.4)
4. **GATE** xpm action(s) of the commit's stage (§13.3); a red gate is fixed by amending, never by a follow-up
5. **TAG** `git tag pr-<ID>` (optional, for bookkeeping)

12.4 Commit message template

```
<type>(<scope>): <imperative summary, at most 72 chars>

Problem:  what is wrong or missing in xpack-development
Failure:  the visible symptom (error text, wrong result, hang)
Change:   what this commit does, file by file if more than one
Scope:    single-core | SMP-guarded | test | build | docs
Gate:     xpm run <action> -> <N>/<N> passed
Group:    <group name> (only for R6 groups)
Source:   golden smp <short sha or subject> | devices@e8e39d5:<path>
Legacy:   PR #<n> of Part I (when it maps to one)

Co-Authored-By: ...
```

type is one of fix, feat, test, build, docs, chore, refactor.

12.5 From commits to pull requests

Each commit is on its own pr/<ID>-<slug> branch, stacked on the previous one in the same repository. A PR targets xpack-development for the first commit and the previous pr/ branch for the others (as in §9.2). After a merge, the next PR is retargeted. Because of R1, R2 and R7, any Phase 1 commit can also be cherry-picked onto xpack-development on its own.

13. Global Ordering, Stages and Gates

13.1 Stages

Stage	Content	Repositories	Ends with gate
S0	Dissolve devices into the arch repos, retire devices	P, C, A32, A64, D	G0
S1	Single-core corrections and improvements (Part I PR #1-#22, plus new ones)	K (+C, P for the hrclock group)	G1 after every commit
S2	SMP core: kernel SMP, POSIX host SMP, Cortex-M SMP contract	K, P, C	G1 after every commit; G2 at the end
S3	New silicon and new architectures: M33, RP2350, AArch32, AArch64 ports	C, A32, A64	Each port commit builds its port library
S4	Test framework: kernel harness, platforms, port test builders, then one commit per test	K, P, C, A32, A64	G1 + G3 (growing)
S5	Documentation, one commit per document	K (+ port READMEs)	None (docs only)
S6	Final parity and the full test run	all	G-final

13.2 Cross-repository order (R6)

```
S0: P01-P03 -> C01-C03 -> A32-01..03 -> A64-01..02 -> D01 (retire devices)
S1: K01 ... K29 (group G-hrclock = K28 + C04 + P04; gate after the last of the three)
S2: K30-K38 (kernel SMP, guarded) -> P05-P11 (host SMP) -> C05-C07 (Cortex-M SMP contract)
S3: C08-C12 (M33, RP2350) | A32-04..12 | A64-04..11 (the three columns are independent)
S4: K40-K47 (harness) -> P20 / C20 / A32-20 / A64-20 (port test builders and boards)
    -> K50.. (one platform per commit) -> one commit per test in each port
S5: K90..
```

13.3 Gates (all of them are xpm actions in micro-os-plus-iii/tests/package.json)

Once per workspace:

```
cd "$WORK/micro-os-plus-iii-posix-arch" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-cortexm" && xpm install && xpm link
cd "$WORK/micro-os-plus-iii-aarch32" && xpm install && xpm link
# from S3
cd "$WORK/micro-os-plus-iii-aarch64" && xpm install && xpm link
# from S3
cd "$WORK/micro-os-plus-iii/tests" && xpm run install-all
```

Gate	Command (in micro-os-plus-iii/tests)	Pass criterion
G0	xpm run test-native-cmake && xpm run test-cortex-cmake	Same result as on xpack-development plus PR #1. Note that the base is red on

Gate	Command (in micro-os-plus-iii/tests)	Pass criterion
		-Werror=extra-semi in dirent.h until K01.
G1	xpm run test-native-cmake + xpm run test-cortex-cmake (the two may run in parallel)	24 configurations × 3 suites = 72/72 with the xpack-development harness. After the harness switch (K44-K46), all ctest entries of the 24 configurations pass, and the new count is written into the Gate: line.
G2	The G1 commands, then the SMP builds of stage S2 through xpm run test-native-cmake (posix-arch test/native/smp-* after P21+)	G1 green, and the SMP host tests pass
G3	xpm run test-<platform>-cmake for the platform the commit touches	All non-hwd tests of the platform pass (ctest -LE hwd)
G-final	xpm run test-all	native + cortex + test-smp-cmake all green
Unifdef review (S2 only, optional)	git diff HEAD~1 unifdef -UOS_USE_SMP_SCHEDULER (review aid)	No single-core change

hwd tests (real hardware: pico2, weact, nucleo, Raspberry Pi, Luckfox) are built by G3 but run only on a bench, through `xpm run test-<...>-hwd --config <cfg>`. Their result is recorded in the commit message when it is available.

14. Stage S0 — Dissolving devices with Commits Only

14.1 Source layout → destination layout

devices@e8e39d5	posix-arch	cortexm	aarch32	aarch64
include/ (sd.hpp, flatfs.hpp, fatfs_hw.hpp, usb_dwc2.hpp)	drivers/ include/	—	drivers/ include/	drivers/ include/
src/ (sd.cpp, flatfs.cpp, usb_dwc2.cpp)	drivers/ src/	—	drivers/ src/	drivers/ src/
fatfs/ (FatFs R0.15 + diskio_sd.cpp, fatfs_hw.cpp)	drivers/ fatfs/	—	drivers/ fatfs/	drivers/ fatfs/
soc/native/	soc/ native/	—	—	—
soc/stm32f4xx/	—	soc/ stm32f4xx/	—	—
soc/rp2350/	—	soc/ rp2350/	—	—
soc/bcm2837/	—	—	soc/ bcm2837/	soc/ bcm2837/
soc/rk3506/ (+ sdcard.md)	—	—	soc/ rk3506/	—

14.2 Commit list

To import without scripts, use `git fetch ../micro-os-plus-iii-devices smp:devices-src` once per arch repo, then `git read-tree --prefix=<dest>/ -u devices-src:<src>` for each row. This only stages the added files; commit each one separately. The branch `devices-src` is local and is deleted after S0.

ID	Repo	Commit	Files	Gate
P01	posix-arch	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers	drivers/{include,src,fatfs}/**	builds (not yet referenced)
P02	posix-arch	feat(soc): import the native host-file SD backend	soc/native/**	idem
P03	posix-arch	build(cmake): define the dissolved devices targets locally	CMakeLists.txt: drop UOS_DEVICES_DIR and its check, add micro-os-plus::devices-hostfile and micro-os-plus::devices (§3.3 block)	G0
C01	cortexm	feat(soc): import the STM32F4xx SoC support	soc/stm32f4xx/**	builds
C02	cortexm	feat(soc): import the RP2350 SoC support	soc/rp2350/**	builds
C03	cortexm	build(cmake): define the dissolved SoC targets locally	CMakeLists.txt: drop UOS_DEVICES_DIR, add soc-stm32f411xe, soc-stm32f412rx, soc-rp2350	G0
A32-01	aarch32	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers	drivers/**	— (the repo has no build yet)
A32-02	aarch32	feat(soc): import the BCM2837 (Raspberry Pi 3 / Zero 2 W) SoC support	soc/bcm2837/**	—
A32-03	aarch32	feat(soc): import the RK3506 (Luckfox Lyra) SoC support	soc/rk3506/**	—
A64-01	aarch64	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers	drivers/**	—
A64-02	aarch64	feat(soc): import the BCM2837 SoC support	soc/bcm2837/**	—
D01	devices	docs: retire the repository -- its contents	README.md only (a table of the moves in §14.1)	—

ID	Repo	Commit	Files	Gate
		moved to the architecture ports		

Each body has the trailer Source: devices@e8e39d5:<src path>. The devices history stays reachable in the archived repository, so a subtree merge is not needed. After D01: archive dan-maio/micro-os-plus-iii-devices on GitHub (Settings → Archive, an administrative action, not a commit), then delete the local clone (rm -rf "\$WORK/micro-os-plus-iii-devices"). From here on, no CMake file may mention micro-os-plus-iii-devices (git grep -n micro-os-plus-iii-devices is empty in all five repositories).

15. Kernel micro-os-plus-iii — Commit List

15.1 Stage S1: single-core corrections and improvements (gate G1 after each)

ID	Commit subject	Files	Legacy
K01	fix(posix): ISO C conformant DIR in dirent.h	include/cmsis-plus/posix/dirent.h	PR #1
K02	fix(utils): list iterators call node_>next()/prev()	include/cmsis-plus/utils/lists.h	PR #2
K03	fix(posix-io): match newlib's syscall return types in the weak aliases	include/cmsis-plus/posix-io/c-syscalls-aliases-standard.h	PR #3
K04	fix(rtos): emit thread::suspend() out of line	src/rtos/os-thread.cpp (that hunk only)	PR #4
K05	fix(libc): declare timegm() only when the C library does not	src/libc/stdlib/timegm.c	new (was hidden in step01)
K06	fix(memory): block_pool asserts on a null result, not a valid one	src/memory/block-pool.cpp	new (was hidden in step02)
K07	fix(estd): polymorphic_allocator copy keeps its resource	include/cmsis-plus/estd/memory_resource	new (was hidden in step02)
K08	feat(memory): usable size for the RTOS allocators, overflow-checked calloc()	rtos/os-memory.h, src/rtos/os-memory.cpp, memory/first-fit-top.h, src/memory/first-fit-top.cpp, src/memory/lifo.cpp, src/libc/stdlib/malloc.cpp	PR #5
K09	feat(libcpp): C++17 aligned operator new/delete	src/libcpp/new.cpp	PR #6
K10	fix(libcpp): error categories with static storage	src/libcpp/system-error.cpp	PR #7
K11	fix(libcpp): steady_clock::now() without 64-bit overflow	src/libcpp/chrono.cpp	PR #8
K12	fix(rtos): CMSIS-RTOS osTimerCreate() defaults to one-shot	src/rtos/os-c-wrapper.cpp (hunk)	PR #9 a

ID	Commit subject	Files	Legacy
K13	fix(rtos): CMSIS-RTOS mutex delete through the polymorphic type	idem (hunk)	PR #9 b
K14	fix(rtos): widen millisec before scaling in the C wrapper	idem (hunk)	PR #9 c
K15	fix(posix-io): serialize the file descriptor table	src/posix-io/file-descriptors-manager.cpp	PR #10
K16	fix(posix-io): lock the deferred file-system and socket lists	posix-io/file-system.h, posix-io/net-stack.h	PR #11
K17	fix(posix-io): block_device sector-size check was inverted	src/posix-io/block-device.cpp	PR #12
K18	feat(arm): ARMv8-M semihosting trap	include/cmsis-plus/arm/semihosting.h	PR #13
K19	feat(startup): ARMv8-M exception handlers and fault dump	src/startup/exception-handlers.c, include/cmsis-plus/cortexm/exception-handlers.h	PR #14
K20	fix(semihosting): fstat() keeps an explicit mode; weak console mirror hook	src/semihosting/c-syscalls-semihosting.cpp	PR #15 (split into two commits if the hunks are independent)
K21	fix(rtos): run timer callbacks outside the critical section	src/rtos/internal/os-lists.cpp, src/rtos/os-timer.cpp (hunks)	PR #16 a
K22	fix(rtos): periodic timer drift catch-up guard	src/rtos/os-timer.cpp (hunks)	PR #16 b
K23	fix(rtos): mutex priority ceiling before ownership, highest boost among waiters	src/rtos/os-mutex.cpp	PR #17
K24	feat(rtos): thread state destroying, and an idle reaper that waits for it	rtos/os-c-decls.h, rtos/os-thread.h, src/rtos/os-idle.cpp, estd/condition_variable (step09 hunks)	PR #18
K25	fix(rtos): condition variable without lost signals	src/rtos/os-condvar.cpp, rtos/os-condvar.h, diag/instrumentation.h (SUSPEND_CAUSE_CONDVAR), estd/condition_variable (step10 hunks)	PR #19
K26	fix(estd): std::thread::join() waits on the native handle, keeps its functor	src/libcpp/thread-cpp.h, estd/thread_internal.h	PR #20
K27	fix(rtos): message queue reschedules after a wake-up	src/rtos/os-mqueue.cpp	PR #21
K28	feat(rtos): high-resolution clock uses the port's hardware counter	rtos/os-decls.h (hunk), src/rtos/os-clocks.cpp	PR #22 K, Group G-hrclock with C04, P04
K29	test(rtos-apis): 512 KiB RTOS arena for the	tests/sources/rtos-apis/include/cmsis-plus/os-app-config.h, tests/sources/	new

ID	Commit subject	Files	Legacy
	FatFs leg, and a switch to leave FatFs out	rtos-apis/src/main.cpp (two commits, K29a/K29b, if you prefer one subject each)	

Also in S1, one commit each, harness corrections that are independent of SMP: | ID | Commit subject | Files | |---|---|---| | K29c | fix(test): enable the FPU when the compiler targets one (__ARM_FP) | tests/device-qemu-cortexm/src/system-cortexm.c | | K29d | docs(test): the AN385 flash comment says 128M | tests/device-qemu-cortexm/linker-scripts/mem-mps2-an385.ld, ...an386.ld |

15.2 Remaining golden kernel changes and their commits (completeness check)

Every kernel file that differs between xpack-development and golden smp belongs to exactly one row of §15.1, §15.3, §15.4, §15.5 or §15.7, or to the intended differences in §19.2. When a file appears in two rows (for example os-thread.cpp in K04 and K35), each row takes only its own hunks.

15.3 Stage S2: kernel SMP core (guarded by OS_USE_SMP_SCHEDULER, gate G1 after each)

ID	Commit subject	Files (SMP hunks only)
K30	feat(port): smp-common -- the SMP declarations a thin port shares	port/smp-common/README.md, port/smp-common/cmsis-plus/ rtos/port/os-decls.h
K31	feat(rtos): SMP_NO_OWNER and the per-CPU thread context	rtos/os-c-decls.h, rtos/os-decls.h
K32	feat(rtos): one current thread per CPU	rtos/os-sched.h
K33	feat(rtos): thread CPU affinity and per-core run state	rtos/os-thread.h
K34	feat(rtos): SMP scheduler -- per-CPU ready lists, recursive kernel lock, no switch while holding it	src/rtos/os-core.cpp
K35	feat(rtos): thread migration and the five-stage stack-pointer claim/publish	src/rtos/os-thread.cpp
K36	feat(rtos): wake a thread's CPU with an IPI when it unblocks elsewhere	src/rtos/os-thread.cpp (IPI hunks; Part I PR #26 kernel part)
K37	feat(rtos): one idle thread per CPU	src/rtos/os-idle.cpp
K38	feat(rtos): pin the main thread to CPU 0	src/rtos/os-main.cpp
K39	test(validator): use glibc's ucontext, not libucontext	tests/sources/cmsis-os-validator/include/cmsis-plus/os-app-config.h (lands right after P05, see R6)

15.4 Stage S4a: build and harness infrastructure (gate G1 after each)

ID	Commit subject	Files
K40	build(cmake): cross toolchain profiles for arm-none-eabi, aarch64-none-elf, native	cmake/toolchains/*.cmake
K41		cmake/uos-app.cmake

ID	Commit subject	Files
	build(cmake): uos-app.cmake -- one call builds an application against a port	
K42	build(cmake): modular kernel targets iiii-core, iiii-posix-io, port-smp-decls, test-support	CMakeLists.txt (keep micro-os-plus::iiii as the umbrella alias)
K42b	build(cmake): fail the configure on a source compiled twice (optional)	tools/verify-no-duplicate-sources.py plus its hook in CMakeLists.txt. If omitted, K42 has no hook.
K43a	test: the verdict contract shared by every board (hw_result, board contract)	test_smpl/include/hw_result.hpp, test_smpl/src/board-contract.cpp
K43b	test: host runner for the port test applications	test_smpl/run-host.sh
K43c	test: QEMU runner for the port test applications	test_smpl/run-qemu.sh
K43d	test: hardware runner and the OpenOCD configs that need no power cycle	test_smpl/run-hw.sh, test_smpl/no-power-cycle/*
K44	test(cmake): find the ports as siblings (UOS_<PORT>_DIR), with the xpack link as fallback	tests/cmake/tests-main.cmake, tests/CMakeLists.txt
K45	test(native): run posix-arch's own test applications	tests/platforms/native/** (after P20, R6)
K46a-d	`test(qemu-cortex-m0	m3
K47	test(qemu): MPS2 AN505/AN521 memory maps and ARMv8-M vectors	tests/device-qemu-cortexm/linker-scripts/mem-mps2-an505.ld, mem-mps2-an521.ld, src/vectors-cortexm.c, include/cmsis_device.h, src/exception-handlers.cpp
K48	test(nucleo-f411re): build against the dissolved soc-stm32f411xe target	tests/platforms/nucleo-f411re/**

15.5 Stage S4b: new test suite (one commit per test)

ID	Commit subject	Files
K49	test: fp-switch -- floating-point context integrity under 1 ms preemption	tests/sources/fp-switch/**

15.6 Stage S4c: platforms (one commit per platform; gate G3 for that platform)

Each commit adds the platform directory **and** in tests/package.json its *-cmake-gcc-{debug,release} configurations, its test-<platform>-cmake action, and **one line** in test-smp-cmake. The first of these commits also creates test-smp-cmake and appends it to test-all. Configuration templates that several platforms share (aarch32-actions, aarch32-dependencies, aarch64-*, arm-cmsis-core-dependencies, clang-gcc14-properties) land with the first platform that uses them.

ID	Platform	Needs (R6)	QEMU in test-smp-cmake
K50	2xcortex-m33 (MPS2 AN521, 2 x M33)	C08, K47	yes
K51		C09	yes

ID	Platform	Needs (R6)	QEMU in test-smp-cmake
	pico2-1cpu (RP2350 code at NCPU=1 on QEMU)		
K52	cortexm-pico2	C21 + pico2 tests	yes
K53	cortexm-pico2-rp2350b-psram	C22	yes
K54	cortexm-pico2-pizero	C23	no (build and hwd only)
K55	cortexm-nucleof411	C24	no
K56	cortexm-weactf411	C25	no
K57	cortexm-weactf412	C26	no
K58	aarch32-rpi-zero-2w (raspi3b, 4 × A53)	A32-20, A32-21	yes
K59	aarch32-rpi3b	A32-22	yes
K60	aarch64-rpi-zero-2w	A64-20, A64-21	yes
K61	aarch64-rpi3b	A64-22	yes
K62	aarch32-luckfox-lyra (RK3506, 3 × A7; hardware only)	A32-23	no (build-aarch32-luckfox-lyra-cmake)
K63	test(package): Debug/Release aliases and the clang 13–15 native configurations	—	—

15.7 Stage S5: documentation (one commit per document)

One commit for each docs/*.md (18 files), for each docs/smp-integration/*.md together with its own diagrams/* (24 documents), for each docs/tests/* group, for docs/specs/*, for README-DEVELOPER.md, tests/README.md and tests/T0-CHECK.md, and for port/smp-common/README.md if it is not in K30. .clang-format and the .gitignore additions are one chore commit each.

16. micro-os-plus-iii-posix-arch — Commit List

16.1 S0 and S1

P01-P03 (§14.2). **P04** feat(port): high-resolution counter from CLOCK_MONOTONIC belongs to group G-hrclock with K28.

16.2 S2: the host as a multi-core machine (one feature per commit; later golden fixes folded in by R5)

ID	Commit subject	Files	Golden fixes folded in
P05	fix(port): header hygiene for GCC < 14 and clang -Weverything	include/cmsis-plus/rtos/port/os-{c-decls,decls,inlines}.h (non-SMP hunks)	"build with GCC < 14 (__has_feature)", Fixes A/B/I
P06	feat(port): SMP port contract -- per-thread CPU,	the same headers (SMP hunks)	"block the tick before reading the CPU id in locked()", "never read

ID	Commit subject	Files	Golden fixes folded in
	per-CPU ISR state, kernel lock		the CPU id through a cached thread pointer"
P07	feat(port): host CPUs are pthreads, with per-CPU timers and signal IPIs	include/host_cpu.hpp, src/host_cpu.cpp, CMakeLists.txt line	"drop the dead timer lock; host CPU tick split", Fixes C/E/G
P08	feat(port): host exception dispatch and the per-core boot stage	include/exception_handler.hpp, src/exception_handler.cpp, CMakeLists.txt line	"handler mode for the validator's SIGUSR1 shim", Fix F
P09	feat(port): the host free store	src/free-store.cpp, CMakeLists.txt line	—
P10	feat(port): SMP start-up and signal-driven context switch	src/rtos/os-core.cpp	"errno is the µOS++ thread's, not the host thread's"
P11	feat(port): the board verdict contract on the host	include/hw_result.hpp, src/board-contract.cpp, CMakeLists.txt line	—
P12	feat(port): sanitizer support for the native board	CMakeLists.txt (hunk)	—
P13	build(cmake): resolve the kernel as a sibling under either directory name	CMakeLists.txt (hunk)	"support micro-os-plus-iii directory name"
P14	tools: a probe showing why TSan fiber annotations cannot work here (optional)	tools/tsan-fiber-probe.c	—

16.3 S4: tests (P20 first, then one commit per test; gate xpm run test-native-cmake)

ID	Commit	Files
P20	test: the port test builder and the native board	test/CMakeLists.txt, test/run.sh, test/boards/native/**, test/native/include/**, test/native/src/**
P21	test(native): rtos-apis at OS_NCPU=1	test/native/rtos-apis/**
P22	test(native): mutex-stress	test/native/mutex-stress/**
P23	test(native): flatfs-test -- append to a file created empty	test/native/flatfs-test/**
P24	test(native): mutex-ceiling-test -- a refused protect lock leaves nothing	test/native/mutex-ceiling-test/**
P25	test(native): smp_test0 ... P29 smp_test4	one directory per commit
P30	test(native): smp-num-test	
P31	test(native): smp-mat-test (N=500, B=50)	

ID	Commit	Files
P32	test(native): smp-pipeline-test	
P33	test(native): smp-pro-cons-test	
P34	test(native): smp-mutex-stress -- SMP leg with per-core checks	
P35	test(native): smp-rtos-apis -- SMP leg with per-core checks	

All of them use `std::atomic` for counters shared between cores (golden fix "convert multi-core volatile counters to `std::atomic`", folded in by R5).

17. micro-os-plus-iii-cortexm — Commit List

17.1 S0 and S1

C01-C03 (§14.2). **C04** `feat(port):` high-resolution counter, with the SysTick pending-overflow check belongs to group G-hrclock.

17.2 S2: Cortex-M SMP contract (single-core Cortex-M builds unchanged)

ID	Commit	Files	Golden fixes folded in
C05	<code>feat(port):</code> Cortex-M SMP contract -- per-core lock state, recursive kernel lock	<code>include/cmsis-plus/rtos/port/os-{c-decls,decls,inlines}.h</code>	"mask IRQs before reading the core id in <code>scheduler::locked()</code> ", "publish scheduler unlock before releasing the lock"
C06	<code>feat(port):</code> PendSV switch with stack-pointer claim/publish	<code>src/rtos/os-core.cpp</code>	"unpublish the stack pointer on an identity reschedule", "release klock on null thread in <code>switch_stacks</code> ", "save the FPU context on a switch"
C07	<code>fix(cortexm):</code> scheduler guards, semihosting clobbers, linker sections	the hunks of that golden commit that touch upstream code	—

17.3 S3: new silicon

ID	Commit	Files
C08	<code>feat(port):</code> generic Cortex-M33 (SSE-200) dual-core port	<code>include-m33/**</code> , <code>src/rtos/os-core-m33.cpp</code> , <code>CMakeLists.txt</code> target
C09	<code>feat(port):</code> RP2350 dual-core port -- SIO spinlock, FIFO IPI, core 1 launch	<code>include-rp2350/**</code> , <code>src/rtos/os-core-rp2350.cpp</code> , <code>CMakeLists.txt</code> target (with "release the 32 SIO spinlocks at reset" folded in)
C10	<code>feat(libc):</code> <code>getentropy()</code> for the bare-metal ports	<code>src/libc/getentropy.c</code> , <code>CMakeLists.txt</code> line
C11	<code>feat:</code> a semihosting exit that reports the test verdict	<code>src/semihosting-exit.cpp</code> , <code>CMakeLists.txt</code> line

ID	Commit	Files
C12	build(cmake): resolve the kernel as a sibling under either directory name	CMakeLists.txt (hunk)

17.4 S4: tests (C20 first, then one board per commit, then one test per commit)

ID	Commit	Files
C20	test: the port test builder, QEMU and hardware dispatch	test/CMakeLists.txt, test/qemu.sh, test/hw.sh, test/boards/shared/**
C21	test(pico2): the Pico 2 board	test/boards/pico2/**, test/pico2/{CMakeLists, tests.cmake, include, src}
C22	test(pico2-rp2350b-psram): the RP2350B PSRAM board	test/boards/pico2-rp2350b-psram/**, test/pico2-rp2350b-psram/tests.cmake
C23	test(pico2-pizero): the Pico 2 on Pi Zero carrier	test/boards/pico2-pizero/**
C24	test(nucleof411): the NUCLE0-F411RE board under the SMP scheduler at one CPU	test/boards/nucleof411/**
C25	test(weactf411): the WeAct F411 board	test/boards/weactf411/**
C26	test(weactf412): the WeAct F412 board	test/boards/weactf412/**

Then one commit per test directory (test(<board>): <test>):

- **pico2**: cmsis-os-validator, cmsis-os-validator-ram, exc-test, fp-switch, mutex-stress, mutex-stress-ram, rtos-apis, rtos-apis-ram, sc-test-ko, smp-mat-test, smp-mat-test-ram, smp-test0 ... smp-test5, smp-test-ko, smp-test-usb-cdc-acm, smp-test-usb-hid (25 commits).
- **pico2-rp2350b-psram**: exc-test, fp-switch, sc-test-ko, smp-mat-test, smp-test0 ... smp-test5, smp-test-ko, smp-test-nested, smp-test-nested-clock, smp-test-nested-clock_200, smp-test-nested-clock_250 (16).
- **pico2-pizero**: exc-test, psram-exec, psram-mat-test-250, sc-test-ko, smp-mat-test, smp-test0 ... smp-test5, smp-test-ko, smp-test-usb-cdc-acm, smp-test-usb-hid (14).
- **nucleof411, weactf411, weactf412**: cmsis-os-validator, mos-test1, mutex-stress, rtos-apis; plus weactf411 spi-pipeline and weactf412 uart-test1 (14).

18. micro-os-plus-iii-aarch32 and micro-os-plus-iii-aarch64 — Commit Lists

On xpack-development both repositories contain only .gitignore, .npmignore, LICENSE, README.md and package.json. Everything is therefore an **add**, plus one .gitignore update.

18.1 S3: the port (each commit builds micro-os-plus::aarch32 / ::aarch64 from the files present so far, R8)

AArch32	AArch64	Commit	Files
A32-04	A64-03	chore: ignore every build* directory	.gitignore

AArch32	AArch64	Commit	Files
A32-05	A64-04	build(cmake): the port project, the dissolved devices target, sibling kernel	CMakeLists.txt (initially only the targets whose sources exist, plus micro-os-plus::devices from §3.3)
A32-06	A64-05	feat(port): device header, application config and port declarations	include/cmsis_device.h, include/cmsis-plus/os-app-config.h, include/cmsis-plus/rtos/port/os-{c-decls,inlines}.h
A32-07	A64-06	feat(port): MMU set-up (A64: with TCR_EL1.EPD1 folded in)	include/mmu.hpp
A32-08	A64-07	feat(port): exception vectors and handlers	include/exception_handler.hpp, src/exception_handler.cpp, src/handlers.cpp
A32-09	A64-08	feat(port): ARM generic timer and the high-resolution counter (CNTPCT / CNTPCT_EL0)	include/timer_arm.hpp
A32-10	A64-09	feat(port): context switch with stack-pointer claim/publish	include/port_ctx.hpp, src/context_switch.cpp, src/rtos/os-core.cpp (MPIDR Aff0 mask folded in)
A32-11	A64-10	feat(port): secondary-core boot	src/smp_secondary.cpp
A32-12	A64-11	feat(port): semihosting console and exit	include/semihosting.hpp, src/semihosting-exit.cpp (A32 only)

18.2 S4: tests

AArch32	AArch64	Commit	Files
A32-20	A64-20	test: the port test builder, QEMU shim and hardware dispatch	test/CMakeLists.txt, test/qemu.sh, test/hw.sh
A32-21	A64-21	test(rpi-zero-2w): the Zero 2 W board	test/boards/rpi-zero-2w/**, test/rpi-zero-2w/{tests.cmake,include,src}
A32-22	A64-22	test(rpi3b): the Pi 3 B board (shares the Zero 2 W sources)	test/boards/rpi3b/**, test/rpi3b/{tests.cmake,include,src}
A32-23	—	test(luckfox-lyra): the Luckfox Lyra board (RK3506, hardware only)	test/boards/luckfox-lyra/**, test/luckfox-lyra/{include,src}

Then one commit per test directory, for each of `rpi-zero-2w` and `rpi3b` in both repositories: `cmsis-os-validator`, `mutex-stress`, `rtos-apis`, `sd_test`, `smp-mat-sdcard-test`, `smp-mat-test`, `smp-num-test`, `smp-pipeline-test`, `smp-pro-cons-test`, `smp_test0` ... `smp_test4`, `usb_test` (18 per board, 72 commits in all).

For `luckfox-lyra` (AArch32 only): `sd_test`, `smp-mat-sdcard-test`, `smp-mat-test`, `smp-num-test`, `smp-pipeline-test`, `smp-pro-cons-test`, `smp_test0` ... `smp_test7`, `smp_test_int`, `smp_test_int2` ... `smp_test_int5` (19 commits).

The kernel platform for a board (K58-K62) lands after the board commit and at least one test. Each further test commit makes one more ctest entry appear in `xpm run test-<platform>-cmake`.

19. Final Parity and the Full Test Run

19.1 Parity check (read-only against golden)

```
for r in micro-os-plus-iii micro-os-plus-iii-cortexm micro-os-plus-
iii-posix-arch \
    micro-os-plus-iii-aarch32 micro-os-plus-iii-aarch64; do
    git -C "$WORK/$r" diff --stat golden/smp smp
done
```

19.2 Intended differences (everything else is a defect to fix with one more commit)

- Files golden deletes but the rebuild keeps (R2): kernel .github/, doxygen/, inspiration/, templates/, .vscode/, .settings/, CHANGELOG.md, CLAUDE.md, README.md, README-MAINTAINER.md, LICENSE (not renamed), package-lock.json, and the link-deps-native-cmake-sys action. Ports: .npmignore, LICENSE, README*.md, CHANGELOG.md, NOTES.md, scripts/xpacks-helper.sh, .vscode/, .clang-format, posix-arch test/trace-*.
smp/**, tests/xpacks/**, README.pdf, and the optional tools/* that were not taken.
- Procedure tooling golden has but the rebuild does not (R3): scripts/smp/**, tests/xpacks/**, README.pdf, and the optional tools/* that were not taken.
- Content the rebuild has but golden lacks: soc/ and drivers/ in posix-arch, cortexm, aarch32 and aarch64. Golden's CMake references them, but they are absent from its tree.
- Any version bump made as an optional chore(release) commit.

19.3 Full test run (µOS++ test framework only)

```
cd "$WORK/micro-os-plus-iii/tests"
xpm run install-all
xpm run test-all # native (gcc11-14, clang16-19) + qemu-
cortex-m0/m3/m4f/m7f + test-smp-cmake
```

test-smp-cmake RUNS aarch32-rpi-zero-2w, aarch32-rpi3b, aarch64-rpi-zero-2w, aarch64-rpi3b, 2xcortex-m33, pico2-1cpu, cortexm-pico2 and cortexm-pico2-rp2350b-psram on QEMU. Hardware-only platforms (cortexm-pico2-pizero, cortexm-nucleof411, cortexm-weact*, aarch32-luckfox-lyra, nucleo-*, raspberrypi-pico) are built by test-all where they are registered, and are run on the bench with their hwd actions.

19.4 Size of the series (planning estimate; the executed counts are in §20.1 and §22.1)

Repository	Commits (approx.)	Of which tests
micro-os-plus-iii	29 + 10 + 13 + 1 + 14 + ~45 docs ≈ 110	1 suite + 13 platforms
posix-arch	3 + 1 + 10 + 16 ≈ 30	16
cortexm	3 + 1 + 3 + 5 + 7 + 69 ≈ 88	69
aarch32	3 + 9 + 4 + 36 + 19 ≈ 71	55
aarch64	2 + 9 + 3 + 36 ≈ 50	36
devices	1 (retirement)	—

Every row is one pull request. Tests and documents are small and add-only, so they review fast and can be merged in bulk once their builder or platform commit has landed.

Part III — Execution Record, Step Summary and Complete Commit List

Part II was executed once from scratch on 2026-10-06. Fresh clones of `github.com/dan-maio/*` went into `/tmp`, with `~/Work2/micro-os-plus/*` as the read-only golden remote. This part records what was done, how it differs from the Part II plan, and every commit that resulted. It is the reference for opening the pull requests.

20. Summary of All Steps: xpack-development (single CPU) → smp (SMP)

20.1 Starting point

Repository	Base (origin/xpack-development)	Result branch	Commits
micro-os-plus-iii (kernel, K)	7f1ce5ca	smp	125 (+ 34 documentation updates, §22.4)
micro-os-plus-iii-posix-arch (P)	86a6a1f	smp	24
micro-os-plus-iii-cortexm (C)	687e975	smp	86
micro-os-plus-iii-aarch32 (A32)	307c62f	smp	64
micro-os-plus-iii-aarch64 (A64)	fe904c0 (GitHub; golden's base 366c957 is one commit older)	smp	43
micro-os-plus-iii-devices (D)	smp	retired (D01)	1
Total			343

Each commit carries the trailer `Commit-ID: <ID>` (`xpack-dev-smp.md` Part II). Each commit also has a local branch `pr/<ID>` pointing at it, so commit *n* becomes the PR `pr/<ID> → pr/<previous ID>` (§12.5). Nothing has been pushed.

20.2 The steps, in the order they were committed

Step	Stage	Repositories and commit IDs	What the step does	Build check
1	Workspace	all	Clone from dan-maio, add the read-only golden remote, create smp from origin/xpack-development (§12.1). Install with xpm <code>install / xpm link</code> in each port, then <code>xpm run install-all</code> in	<code>install-all</code> <code>rc 0</code>

Step	Stage	Repositories and commit IDs	What the step does	Build check
			micro-os-plus-iii/tests (\$13.3).	
2	S0 devices	P01-P03, C01-C03, A32-01..03, A64-01..02, D01	Dissolve devices. Its drivers (SD, flatfs, USB DWC2, FatFs) go to drivers/ and its SoC support to soc/<chip>/ in each port that uses them. The micro-os-plus::devices* / soc-* CMake targets are defined in the port. The devices repo gets one README commit that records where everything went.	native + cortex build
3	S1 single-core	K01-K29d (41 commits)	Upstream corrections and improvements, one per commit. They cover dirent, lists, the syscall aliases, suspend, the allocators, aligned new, system_error, chrono, the C API, POSIX I/O locking, block device, ARMv8-M semihosting and fault handlers, timers, the mutex ceiling and inheritance, the thread life cycle (destroying, detach, join), the condition variable, std::thread, mqueue, and the rtos-apis test arena.	build after each
4	S1 hrclock group	K28a, K28 + C04, C04b + P04	The high-resolution clock reads the port's hardware counter. The kernel declares it, cortexm reports that it has no hardware counter and fixes cycles_since_tick(), and posix-arch reads CLOCK_MONOTONIC.	builds at the last commit of the group
5	S2 SMP core (kernel)	K30-K39b	Everything is guarded by OS_USE_SMP_SCHEDULER: port/smp-common, one current thread per CPU, CPU affinity, the SMP picker with the stack-pointer claim, the IPI wake-up, join/kill/ reaper aware of other CPUs, main thread and CMSIS-RTOS v1 threads on CPU 0.	single-core builds unchanged
6	S2 host SMP	P05, P06	posix-arch becomes a multi-core host (one host thread per CPU, signal IPIs, per-CPU timers, exception dispatch) and gets the board verdict contract.	native build
7		C05-C08		

Step	Stage	Repositories and commit IDs	What the step does	Build check
	S2 Cortex-M SMP		The SMP port contract on the generic Cortex-M core (SMP scheduler at one CPU), a <code>_getentropy()</code> stub, and a semihosting <code>_Exit()</code> that reports the verdict.	cortex build
8	S3 new silicon	C09, C10	A generic dual-core Cortex-M33 (SSE-200) port for QEMU, and the RP2350 dual-core port.	build
9	S3 new architectures	A32-04..11, A64-03..10	The AArch32 and AArch64 ports: declarations, MMU, vectors, generic timer, context switch, secondary-core entry, semihosting. Add-only; nothing builds them until step 12.	— (no board yet)
10	S4a harness	K40, K41, K43a-d, then group G-harness: K42, K44, K45, K46-m0..m7f + P20 + C20a/b, then K47	Toolchain profiles, <code>uos_add_app()</code> , the <code>test_smpl</code> verdict contract and runners, and modular kernel targets. Each port has a test builder that selects a board, and the kernel harness finds the ports as siblings. The native and QEMU Cortex-M platforms build against the port targets.	builds at the last commit of the group
11	S4b/c Cortex-M platforms	K49, K50-K57; C21-C27 boards; one commit per test	The fp-switch suite; platforms <code>2xcortex-m33</code> , <code>pico2-1cpu</code> , <code>cortexm-pico2</code> , <code>-rp2350b-psram</code> , <code>-pizero</code> , <code>-nucleof411</code> , <code>-weactf411</code> , <code>-weactf412</code> ; the boards <code>pico2</code> (with TinyUSB in C21b-d), <code>pico2-rp2350b-psram</code> , <code>pico2-pizero</code> , <code>nucleof411</code> , <code>weactf411</code> and <code>weactf412</code> ; 63 test commits.	build per commit
12	S4 AArch32/AArch64 boards	A32-20, A32-21, A32-R3, A32-LL; A64-20, A64-21, A64-R3; K58-K62	Port CMake, board models (<code>rpi-zero-2w</code> , <code>rpi3b</code> , <code>luckfox-lyra</code>), and the kernel platforms <code>aarch32-rpi-zero-2w</code> , <code>aarch32-rpi3b</code> , <code>aarch32-luckfox-lyra</code> , <code>aarch64-rpi-zero-2w</code> , <code>aarch64-rpi3b</code> .	build per commit
13	S4 port tests	P21-P35, A32-T-, A32-R3-, A32-LL-, A64-T-, A64-R3-*	One commit per test directory, each adding one <code>ctest</code> entry: 15 native, 30 AArch32 Raspberry Pi, 20 Luckfox Lyra, 30 AArch64.	build per commit
14		K39c, K48-nucleo-	The validator host build uses glibc's <code>ucontext</code> .	see §21.3

Step	Stage	Repositories and commit IDs	What the step does	Build check
	S4 remaining harness	f411re/-f767zi/-h743zi/-raspberrypi-pico	The legacy hardware platforms label their tests hwd and link the modular targets.	
15	S5 docs and chores	K901–K935, K980–K984, P14, P36, C28, C29; after the run K985–K989 (§22.4)	One commit per document; .clang-format; .gitignore additions; the TSan probe; the cortexm board README; this document.	—
16	S6 parity	all (read-only diff with golden/smp)	§21.4.	—
17	S6 full test	micro-os-plus-iii/tests	§21.5.	all green

21. Execution Record

21.1 Where the executed IDs differ from the Part II plan

Plan (Part II)	Executed	Reason
K05 <code>timegm()</code> declaration	dropped	Golden's version fails on glibc 2.44 with <code>-Werror=missing-prototypes</code> . The baseline <code>timegm.c</code> is kept (intended difference).
K20, K23, K24, K25, K26	split into K20a/b, K23a–c, K24a–e, K25a/b, K26a/b	R1: their golden hunks turned out to hold several independent subjects.
K21a (new)	<code>fix(rtos): errno in handler mode does not assert</code>	A separate golden subject in the K21 files.
K22 periodic-timer drift guard	folded into K21	It shares one hunk with the relink (stated in K21's Legacy: line).
K28a (new)	<code>os-decls.h</code> includes the C declarations again	The provider of K28 (R6).
K31–K39	renumbered by subject (see §22.2)	The golden SMP hunks group by behaviour (current thread, affinity, picker, IPI, join, kill, reaper, CPU 0) rather than by file.
K39b (new)	<code>port_cpu_id()</code> declarations tolerate the port's own	Needed by P05.
K42b duplicate-source check	not taken	It is a Python tool (R3).
K46a–d	K46-m0, -m3, -m4f, -m7f	Named by platform.
K48	four commits K48-nucleo-f411re/-f767zi/-h743zi/-raspberrypi-pico	One per legacy platform (R1).
K63 Debug/Release aliases, clang 13–15	not needed	<code>tests/package.json</code> is already byte-identical to golden after K62.

Plan (Part II)	Executed	Reason
K90.. docs	K901-K935 (one per document) + K980-K984	§15.7.
P05-P13 host SMP	P05 (one commit) + P06	Golden's host-SMP files only compile together. The fixes (A-J, GCC < 14, errno) are folded in (R5).
C05-C12	C04b, C05-C10	<code>cycles_since_tick()</code> became its own fix. <code>getentropy/_Exit</code> came before the new silicon. The CMake sibling lookup was folded into C20b.
C20	C20a (nucleof411 board, needed by the builder) + C20b (builder)	R6.
C21-C26 boards	C21 (+C21b-d TinyUSB), C23-C27	pico2's USB stack is three commits. The pico2 test series is C22-*.
A32-05..12 / A64-04..11	A32-05..11 / A64-04..10, then A32-21 / A64-21 for CMake	Port CMake lands with the first board (R8: no target without sources).
A32-22/23, A64-22	A32-R3, A32-LL, A64-R3	Named by board.

21.2 Groups (built only at their last commit)

Group	Commits	Why
G-hrclk	K28a, K28, C04, C04b, P04	The kernel calls <code>clock_highres::has_hardware_counter()</code> , which each port defines.
G-harness	K42, K44, K45, K46-m0..m7f, P20, C20a, C20b	The kernel test harness switches from its own device sources to the port test builders, and neither side builds alone. The messages of the intermediate commits say so.

21.3 Builds

Every commit was built with `xpm run build --config <cfg>` in `micro-os-plus-iii/tests`, with uncommitted changes stashed, and was green, except:

1. The intermediate commits of the two groups (§21.2).
2. A32-01..11 and A64-01..10. They only add files, and nothing builds them until the board commits A32-21 / A64-21.
3. `nucleo-f411re`, `nucleo-f767zi`, `nucleo-h743zi` and `raspberrypi-pico` fail at configure: the golden `cortexm` CMake does not find the kernel through the `xpacks` path. The files are identical to golden, so this is a golden defect, reported here and not fixed in this series.

Build fixes beyond golden, each folded into the commit that introduced the code (R5):

- warning suppressions in `system-error.cpp`, `first-fit-top.cpp`, `file-descriptors-manager.cpp`;
- Fix `H` pragmas in `os-core.cpp` and `os-thread.cpp`;
- the kernel test-support target falls back to `test_smpl/src/board-contract.cpp` (K42);
- `posix-arch` fixes A-J (P05);
- the `AArch32 micro-os-plus::soc-bcm2837` and `micro-os-plus::devices-rk3506` targets (A32-21);
- the `AArch64 soc-bcm2837` target (A64-21).

21.4 Final parity with ~/Work2/micro-os-plus/* (golden/smp, devices excluded)

- **Identical:** all cortexm sources and CMake, and tests/package.json.
- **Kept although golden deletes them (R2):** .github/, doxygen/, inspiration/, templates/, .vscode/, .settings/, CHANGELOG.md, CLAUDE.md, README*.md, LICENSE (not renamed), package-lock.json, the port LICENSE/.npmignore/README.md, posix-arch test/trace-*, and scripts/xpacks-helper.sh.
- **Golden content not taken (R3):** scripts/smp/**, tools/verify-*, docs/**/*.*sh and *.py, the diagram generators, tests/xpacks/**.
- **In the rebuild but not in golden:** soc/ and drivers/ in all four ports (the dissolved devices).
- **Build fixes:** listed in §21.3; plus timegm.c at baseline.
- **Other:** the aarch64 package.json description is the original one, and this document's Part II and Part III are new.

No other difference remains.

21.5 Full test run (µOS++ test framework, latest compilers, sys for native)

Action (in micro-os-plus-iii/tests)	Result
xpm run install-all	rc 0
xpm run test-native-cmake-sys (debug, release)	16/16 passed in each
xpm run test-cortex-cmake (qemu-cortex-m0/m3/m4f/m7f × gcc/clang × debug/release = 8 configs)	3/3 passed in each. The CMSIS validator reports 60/60.
xpm run test-smp-cmake: aarch32-rpi-zero-2w, aarch32-rpi3b (debug, release)	15/15 passed in each
aarch64-rpi-zero-2w, aarch64-rpi3b (debug, release)	15/15 passed in each
2xcortex-m33 (debug, release)	4/4 passed in each
pico2-1cpu (debug, release)	4/4 passed in each
cortexm-pico2 (debug, release)	6/6 passed in each
cortexm-pico2-rp2350b-psram (debug, release)	3/3 passed in each

All of them run on QEMU or on the host; ctest -LE hwd excludes the real-hardware tests. The hardware-only platforms are built but not run.

21.6 Verification replay from golden (2026-10-06)

To check that the procedure of Part II, as recorded here, rebuilds the smp branches, the whole series was executed again from scratch in an empty /tmp:

1. **Clone.** Each repository was cloned from the golden working copies in ~/Work2/micro-os-plus/, branch xpack-development, with the documented bases: kernel 7f1ce5ca, cortexm 687e975, posix-arch 86a6a1f, aarch32 307c62f, aarch64 fe904c0, devices e8e39d5. The golden repositories are the read-only golden remote; the kernel's golden/smp is the commit before this document's own golden commit.
2. **Install.** xpm run install-all in micro-os-plus-iii/tests, then xpm install and xpm link in each port (§13.3).
3. **Commits.** Every commit of §22 was made again in the same order, with the same lifts and hunk selections, and built after each commit as in the first run (xpm run build --config <cfg>).
4. **Result.** The same 375 commits (kernel 158, posix-arch 24, cortexm 86, aarch32 64, aarch64 43) plus devices D01, with the same Commit-IDs, order and subjects, and the same build result at every

commit (the exceptions of §21.3 included). Only the SHAs differ, because the commit dates differ; the SHAs in §22.3 are those of this replay.

5. **Parity with golden smp.** Exactly the intended differences of §21.4, plus the documentation updates of §22.4. This document itself differs from the golden copy only in the SHA columns.
6. **Full test run** (latest compilers, `cctest -LE hwd`):

Action	Result
xpm run install-all	rc 0
xpm run test- native-cmake- sys	16/16 in debug and release
xpm run test- cortex-cmake	3/3 in each of the 8 configurations
xpm run test- smp-cmake	aarch32 and aarch64 rpi-zero-2w/rpi3b 15/15, 2xcortex-m33 4/4, pico2-1cpu 4/4, cortexm-pico2 6/6, cortexm-pico2-rp2350b-psram 3/3 — debug and release, rc 0

In the first pass of test-smp-cmake, aarch32-rpi-zero-2w-smp-pipeline-test-qemu (release) stopped printing after 26 s of guest time and was killed at its 1000 s timeout. Run alone it then passed 5 times in 5 (35 s each), and the complete second pass of test-smp-cmake was green. It is recorded as an intermittent stall of the SMP pipeline test under QEMU (raspi3b, 4 cores), to be investigated separately; it is not a defect of the migration series.

22. Complete Pull-Request and Commit List (all repositories)

Every commit of every repository is one pull request. A PR's head is the local branch `pr/<Commit-ID>`, which points at that commit; its base is the previous PR's branch in the same repository (the first PR of a repository targets `xpack-development`). After a PR is merged, the next one is retargeted to `xpack-development` (§12.5). The branches exist locally in each `/tmp/<repository>` clone and are pushed to `github.com/dan-maio/<repository>` only when the PRs are opened.

22.1 Legend and totals

- **PR**: position on `smp`, counted from the base (§20.1); this is the order in which the repository's PRs are opened and merged.
- **Head branch (Commit-ID)**: `pr/<ID>`, where `<ID>` is the ID in the commit trailer `Commit-ID: <ID>` (`xpack-dev-smp.md` Part II).
- **Base branch**: the branch the PR targets.
- **Category**: correction (fix), new code (feat), test, build, docs, chore, tool.
- **Stage / theme**: the Part II stage (§13.1), followed by the R6 group (§21.2) when the commit belongs to one.

Repository	Corrections	New code	Tests	Build	Docs	Chore / tool	PRs
micro-os-plus-iii (migration)	31	18	32	3	39	2	125
micro-os-plus-iii (documentation updates, §22.4)	0	0	0	0	34	0	34
micro-os-plus-iii-posix-arch	0	5	15	2	0	2	24
micro-os-plus-iii-cortexm	1	9	72	2	1	1	86

Repository	Corrections	New code	Tests	Build	Docs	Chore / tool	PRs
micro-os-plus-iii-aarch32	0	10	52	1	0	1	64
micro-os-plus-iii-aarch64	0	9	32	1	0	1	43
micro-os-plus-iii-devices	0	0	0	0	1	0	1
Total	32	51	203	9	75	7	377

22.2 Cross-repository merge order

The PRs of one repository merge in their own order (22.3). Across repositories, R6 (provider before consumer) adds these constraints:

Before	Merge	Why
—	S0 in every port: posix-arch PR 1-3, cortexm PR 1-3, aarch32 PR 1-3, aarch64 PR 1-2; then devices PR 1	The drivers and SoC support exist in the ports before the devices repository is retired.
kernel K28a	G-hrclock together: kernel K28 + cortexm C04, C04b + posix-arch P04	The kernel calls <code>clock_highres::has_hardware_counter()</code> , which each port defines.
kernel K39b	posix-arch P05, P06	The host SMP port relies on the kernel's tolerant <code>port_cpu_id()</code> declarations.
kernel K30-K39b, K40, K41, K43a-d	G-harness together: kernel K42, K44, K45, K46-m0..m7f + posix-arch P20 + cortexm C20a, C20b	The kernel harness switches to the port test builders; neither side builds alone.
cortexm C09 / C10 and its board PR	kernel K50 (2xcortex-m33), K51 (pico2-1cpu), K52-K57 (one per cortexm board)	A kernel platform needs its port core and board.
aarch32 A32-20, A32-21 (and A32-R3, A32-LL)	kernel K58, K59, K62	idem for AArch32.
aarch64 A64-20, A64-21 (and A64-R3)	kernel K60, K61	idem for AArch64.
the board PR of a port	that board's test PRs (pr/<board-ID>-<test>)	One test directory = one ctest entry on an existing platform.

22.3 Pull requests per repository

micro-os-plus-iii -- 159 pull requests (github.com/dan-maio/micro-os-plus-iii)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/K01	xpack-development	b364e689	correction	S1 single-core	fix(posix): ISO C conformant DIR in dirent.h
2	pr/K02	pr/K01	52a7f499	correction	S1 single-core	fix(utils): list iterators call node_>next()/prev()
3	pr/K03	pr/K02	2f8c482e	correction	S1 single-core	fix(posix-io): match newlib's syscall return types in the weak aliases
4	pr/K04	pr/K03	bcb5737e	correction	S1 single-core	fix(rtos): emit this_thread::suspend() out of line
5	pr/K06	pr/K04	ef9bb2ce	correction	S1 single-core	fix(memory): block_pool asserts on a null result, not on a valid one
6	pr/K07	pr/K06	a6d6c475	correction	S1 single-core	fix(estd): polymorphic_allocator copy keeps its memory resource
7	pr/K08	pr/K07	51bdad10	new code	S1 single-core	feat(memory): usable size for the RTOS allocators, overflow-checked calloc()
8	pr/K09	pr/K08	1c589065	new code	S1 single-core	feat(libcpp): C++17 aligned operator new/delete
9	pr/K10	pr/K09	de47254b	correction	S1 single-core	fix(libcpp): error categories with static storage
10	pr/K11	pr/K10	c8aa3024	correction	S1 single-core	fix(libcpp): steady_clock::now() without 64-bit overflow
11	pr/K12	pr/K11	0c11e258	correction	S1 single-core	fix(rtos): C API timers default to one-shot
12	pr/K13	pr/K12	ebfa07b3	correction	S1 single-core	fix(rtos): C API deletes mutexes and semaphores through their concrete type
13	pr/K14	pr/K13	b4d6fdcf	correction	S1 single-core	fix(rtos): widen millisec before scaling in the CMSIS-RTOS API
14	pr/K15	pr/K14	8ed1c7b0	correction	S1 single-core	fix(posix-io): serialize the file descriptor table
15	pr/K16	pr/K15	fa232ed7	correction	S1 single-core	fix(posix-io): lock the deferred file-system and socket lists
16	pr/K17	pr/K16	01b50480	correction	S1 single-core	fix(posix-io): block_device size queries tested the wrong condition
17	pr/K18	pr/K17	deebab3	new code	S1 single-core	feat(arm): semihosting trap for ARMv8-M, and an optional HLT trap
18	pr/K19	pr/K18	b85b378a	new code	S1 single-core	feat(startup): exception handlers and fault dump for ARMv8-M Mainline

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
19	pr/K20a	pr/K19	182b5f41	correction	S1 single- core	fix(semihosting): fstat() keeps a file type the caller already set
20	pr/K20b	pr/K20a	98eaf96f	new code	S1 single- core	feat(semihosting): weak board console mirror for stdout/ stderr
21	pr/K21a	pr/K20b	236b2d1a	correction	S1 single- core	fix(rtos): errno in handler mode does not assert
22	pr/K21	pr/K21a	8ea2f2f6	correction	S1 single- core	fix(rtos): run timer callbacks outside the critical section
23	pr/K23a	pr/K21	267b955b	correction	S1 single- core	fix(rtos): refuse a priority- ceiling lock before taking ownership
24	pr/K23b	pr/K23a	f5861fc2	correction	S1 single- core	fix(rtos): priority inheritance keeps the highest boost among waiters
25	pr/K23c	pr/K23b	e2b3709c	correction	S1 single- core	fix(rtos): mutex lock re-checks the owner after boosting it
26	pr/K24a	pr/K23c	5390f31d	new code	S1 single- core	feat(rtos): thread state destroying, claimed by exactly one destroyer
27	pr/K24b	pr/K24a	8fcfbc41	correction	S1 single- core	fix(rtos): resume() re-enqueues only a suspended or initializing thread
28	pr/K24c	pr/K24b	61756303	correction	S1 single- core	fix(rtos): priority changes test and relink under one lock
29	pr/K24d	pr/K24c	94dce976	new code	S1 single- core	feat(rtos): implement thread::detach()
30	pr/K24e	pr/K24d	af3dfd7b	correction	S1 single- core	fix(rtos): join() and thread destruction without a lost wake-up
31	pr/K25a	pr/K24e	325a75cc	correction	S1 single- core	fix(estd): condition_variable::wait_for() trusts the kernel's timeout verdict
32	pr/K25b	pr/K25a	102a9136	correction	S1 single- core	fix(rtos): condition variable without lost signals, bound to its clock
33	pr/K26a	pr/K25b	e57aff39	correction	S1 single- core	fix(estd): std::thread keeps its own pointer to the bound function object
34	pr/K26b	pr/K26a	8742e5f4	correction	S1 single- core	fix(estd): std::thread::join() waits for the thread before freeing it
35	pr/K27	pr/K26b	72af88d8	correction	S1 single- core	fix(rtos): message queue reschedules after waking a peer
36	pr/K28a	pr/K27	135edeb3	correction	S1 single- core	fix(rtos): <cmsis-plus/rtos/os- decls.h> includes the C declarations again
37	pr/K28	pr/K28a	b474cf72	new code	S1 single- core	feat(rtos): the high-resolution clock reads the port's hardware counter

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
					G-hr clock	
38	pr/K29a	pr/K28	4804b93c	test	S1 single- core	test(rtos-apis): a 512 KiB RTOS arena
39	pr/K29b	pr/K29a	675a698e	test	S1 single- core	test(rtos-apis): OS_EXCLUDE_RTOS_APIS_FATFS leaves the FatFs leg out
40	pr/K29c	pr/K29b	f242e451	correction	S1 single- core	fix(test): enable the FPU when the compiler targets one
41	pr/K29d	pr/K29c	9cb742d0	docs	S1 single- core	docs(test): the MPS2 AN385/AN386 flash comment says 128M
42	pr/K30	pr/K29d	6f51da7a	new code	S2 SMP core	feat(port): smp-common -- the SMP declarations a thin port shares
43	pr/K31	pr/K30	f1894152	new code	S2 SMP core	feat(rtos): one current thread per CPU
44	pr/K32	pr/K31	604321a6	new code	S2 SMP core	feat(rtos): thread CPU affinity
45	pr/K33	pr/K32	ccfc03ec	new code	S2 SMP core	feat(rtos): SMP thread picker with affinity and the stack-pointer claim
46	pr/K34	pr/K33	3764d613	new code	S2 SMP core	feat(rtos): wake an eligible CPU with an IPI when a thread is resumed elsewhere
47	pr/K35	pr/K34	ce44e5b8	new code	S2 SMP core	feat(rtos): join() returns only once the thread is off every CPU
48	pr/K36	pr/K35	1a811b99	new code	S2 SMP core	feat(rtos): kill() waits until the thread is off every CPU and unclaimed
49	pr/K37	pr/K36	a5194f6e	new code	S2 SMP core	feat(rtos): the idle reaper leaves a thread that is still live on another CPU
50	pr/K38	pr/K37	bcab09e7	new code	S2 SMP core	feat(rtos): pin the main thread to CPU 0
51	pr/K39	pr/K38	b4e819eb	new code	S2 SMP core	feat(rtos): CMSIS-RTOS v1 threads run on CPU 0
52	pr/K39b	pr/K39	12882e19	correction	S2 SMP core	fix(rtos): the kernel's port_cpu_id() declarations tolerate the port's own
53	pr/K40	pr/K39b	1f6d219f	build	S4a harness	build(cmake): toolchain profiles for arm-none-eabi, aarch64-none-elf and the host
54	pr/K41	pr/K40	68155aee	build	S4a harness	build(cmake): uos_add_app() -- one call declares an application against a port
55	pr/K43a	pr/K41	0b6d9339	test	S4a harness	test: the verdict contract every board shares (hw_result, board contract)
56	pr/K43b	pr/K43a	a569c214	test	S4a harness	test: the host runner for the port test applications
57	pr/K43c	pr/K43b	4db26232	test	S4a harness	test: the QEMU runner for the port test applications
58	pr/K43d	pr/K43c	0092d31d	test	S4a harness	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						test: the hardware runner and the OpenOCD configs that need no power cycle
59	pr/K42	pr/K43d	a530fac4	build	S4a harness · G- harness	build(cmake): modular kernel targets
60	pr/K44	pr/K42	2f21e8ca	test	S4a harness · G- harness	test(cmake): the library under test is the architecture port, found as a sibling
61	pr/K45	pr/K44	5fef63ae	test	S4a harness · G- harness	test(native): run posix-arch's own test applications
62	pr/K46-m0	pr/K45	62d55a6e	test	S4a harness · G- harness	test(qemu-cortex-m0): build against the cortexm port's generic QEMU core
63	pr/K46-m3	pr/K46-m0	acc3e284	test	S4a harness · G- harness	test(qemu-cortex-m3): build against the cortexm port's generic QEMU core
64	pr/K46-m4f	pr/K46-m3	b3322cc3	test	S4a harness · G- harness	test(qemu-cortex-m4f): build against the cortexm port's generic QEMU core
65	pr/K46-m7f	pr/K46-m4f	d0edbce3	test	S4a harness · G- harness	test(qemu-cortex-m7f): build against the cortexm port's generic QEMU core
66	pr/K47	pr/K46-m7f	c91f3b6b	test	S4a harness	test(qemu): MPS2 AN505/AN521 memory maps and ARMv8-M support in the QEMU device
67	pr/K49	pr/K47	9a194e45	test	S4b test suite	test: fp-switch -- the FPU registers survive a preemptive switch and a core move
68	pr/K50	pr/K49	8efc3ecb	test	S4c platform	test(2xcortex-m33): two Cortex-M33 cores on QEMU mps2-an521
69	pr/K51	pr/K50	85c38290	test	S4c platform	test(pico2-1cpu): the RP2350's Cortex-M33 code at one CPU on QEMU
70	pr/K52	pr/K51	5af5b8d7	test	S4c platform	test(cortexm-pico2): the Pico 2 board's own tests, on QEMU and on the board
71	pr/K53	pr/K52	41ee57a4	test	S4c platform	test(cortexm-pico2-rp2350b-psram): the RP2350B PSRAM board's tests
72	pr/K54	pr/K53	e5a581a8	test	S4c platform	test(cortexm-pico2-pizero): the pico2-pizero board's tests, hardware only
73	pr/K55	pr/K54	76fd013c	test	S4c platform	test(cortexm-nucleof411): the NUCLE0-F411RE board's tests under the SMP scheduler at one CPU
74	pr/K56	pr/K55	c0772ab8	test	S4c platform	test(cortexm-weactf411): the WeAct F411 board's tests
75	pr/K57	pr/K56	084b6528	test	S4c platform	test(cortexm-weactf412): the WeAct F412 board's tests
76	pr/K58	pr/K57	93faa7da	test		

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
					S4c platform	test(aarch32-rpi-zero-2w): the Zero 2 W's AArch32 tests on QEMU raspi3b
77	pr/K59	pr/K58	8f3a1051	test	S4c platform	test(aarch32-rpi3b): the Pi 3 B's AArch32 tests on QEMU raspi3b
78	pr/K62	pr/K59	9ee32a05	test	S4c platform	test(aarch32-luckfox-lyra): build the Luckfox Lyra's tests
79	pr/K60	pr/K62	55841a50	test	S4c platform	test(aarch64-rpi-zero-2w): the Zero 2 W's AArch64 tests on QEMU raspi3b
80	pr/K61	pr/K60	eb23b89a	test	S4c platform	test(aarch64-rpi3b): the Pi 3 B's AArch64 tests on QEMU raspi3b
81	pr/K39c	pr/K61	c329210b	test	S2 SMP core	test(validator): the host build uses glibc's ucontext, not libucontext
82	pr/K48-nucleo- f411re	pr/K39c	a62e6992	test	S4a harness	test(nucleo-f411re): label the board tests hwd and link the kernel's modular targets
83	pr/K48-nucleo- f767zi	pr/K48-nucleo- f411re	0f3d65ad	test	S4a harness	test(nucleo-f767zi): label the board tests hwd and link the kernel's modular targets
84	pr/K48-nucleo- h743zi	pr/K48-nucleo- f767zi	cfa25b4c	test	S4a harness	test(nucleo-h743zi): label the board tests hwd and link the kernel's modular targets
85	pr/K48- raspberrypi-pico	pr/K48-nucleo- h743zi	87be7f54	test	S4a harness	test(raspberrypi-pico): label the board tests hwd and link the kernel's modular targets
86	pr/K901	pr/K48- raspberrypi-pico	eb6fb4d9	docs	S5 docs/ chore	docs: aarch32-second-board
87	pr/K902	pr/K901	b8c45608	docs	S5 docs/ chore	docs: building-aarch32-aarch64
88	pr/K903	pr/K902	5f899167	docs	S5 docs/ chore	docs: cortexm-port
89	pr/K904	pr/K903	8800c1c7	docs	S5 docs/ chore	docs: posix-arch-port
90	pr/K905	pr/K904	c3f5b83a	docs	S5 docs/ chore	docs: smp-integration/agy- review
91	pr/K906	pr/K905	b2027337	docs	S5 docs/ chore	docs: smp-integration/DeepSeek- review
92	pr/K907	pr/K906	c98fcc93	docs	S5 docs/ chore	docs: smp-integration/diagrams
93	pr/K908	pr/K907	8ddcb6c6	docs	S5 docs/ chore	docs: smp-integration/files- modif-by-step
94	pr/K909	pr/K908	b9695be3	docs	S5 docs/ chore	docs: smp-integration/GITHUB- PROGRESSIVE-PR-GUIDE
95	pr/K910	pr/K909	ac125125	docs	S5 docs/ chore	docs: smp-integration/ Implementation-SMP-Integration
96	pr/K911	pr/K910	75e7e1c3	docs	S5 docs/ chore	docs: smp-integration/micro-os- plus-iii-project-unification
97	pr/K912	pr/K911	0e26f6e2	docs	S5 docs/ chore	docs: smp-integration/MICRO-OS- PLUS-SMP-VS-SINGLECORE-ANALYSIS
98	pr/K913	pr/K912	371d71b1	docs	S5 docs/ chore	docs: smp-integration/new- modifications
99	pr/K914	pr/K913	8930f3c7	docs		

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
					S5 docs/ chore	docs: smp-integration/pull-request
100	pr/K915	pr/K914	6658ffb3	docs	S5 docs/ chore	docs: smp-integration/smp-construction
101	pr/K916	pr/K915	28e2fb7d	docs	S5 docs/ chore	docs: smp-integration/SMP-UPSTREAM-INTEGRATION-PLAN
102	pr/K917	pr/K916	a840dec4	docs	S5 docs/ chore	docs: smp-integration/xpack-dev-smp
103	pr/K918	pr/K917	8fdb54ca	docs	S5 docs/ chore	docs: specs
104	pr/K919	pr/K918	32f6644e	docs	S5 docs/ chore	docs: STATUS
105	pr/K920	pr/K919	5e847731	docs	S5 docs/ chore	docs: tests/AARCH32-RPI-ZERO-2W-TESTS
106	pr/K921	pr/K920	d925dc2c	docs	S5 docs/ chore	docs: tests/HARNESS-BOARD-TEST-CHEATSHEET
107	pr/K922	pr/K921	bc6c523b	docs	S5 docs/ chore	docs: tests/HARNESS-TESTS-PARADIGM
108	pr/K923	pr/K922	8c4fbd55	docs	S5 docs/ chore	docs: tests-in-aarch32-aarch64
109	pr/K924	pr/K923	646cb078	docs	S5 docs/ chore	docs: test-smpl
110	pr/K925	pr/K924	a136e8bd	docs	S5 docs/ chore	docs: tests/README
111	pr/K926	pr/K925	20bf4b20	docs	S5 docs/ chore	docs: tests/README-DEVELOPER
112	pr/K927	pr/K926	9b8d0fe9	docs	S5 docs/ chore	docs: tests/README-MAINTAINER
113	pr/K928	pr/K927	d53e34c7	docs	S5 docs/ chore	docs: tests/STEPS
114	pr/K929	pr/K928	3a0b2153	docs	S5 docs/ chore	docs: tests/test-framework
115	pr/K930	pr/K929	4815480b	docs	S5 docs/ chore	docs: tests/TESTS-CATALOG
116	pr/K931	pr/K930	d142d5a1	docs	S5 docs/ chore	docs: tests/TESTS-DEVELOPER-GUIDE
117	pr/K932	pr/K931	c72b7500	docs	S5 docs/ chore	docs: tests/TESTS-XPack-SYSTEM
118	pr/K933	pr/K932	1deca033	docs	S5 docs/ chore	docs: tests/WORK-SMP-AARCH32-AARCH64-HARNESS-GUIDE
119	pr/K934	pr/K933	72f5d7a1	docs	S5 docs/ chore	docs: upstream-CHANGELOG
120	pr/K935	pr/K934	6e0c1276	docs	S5 docs/ chore	docs: upstream-package
121	pr/K980	pr/K935	675652cb	docs	S5 docs/ chore	docs: README-DEVELOPER updated from the reference smp branch
122	pr/K981	pr/K980	f6337d83	docs	S5 docs/ chore	docs(test): tests/README for the 22 platforms, T0-CHECK marked as history
123	pr/K982	pr/K981	0301394d	chore	S5 docs/ chore	chore: a .clang-format for the kernel sources
124	pr/K983	pr/K982	c7541e17	chore	S5 docs/ chore	chore: ignore every build* directory and the build products

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
125	pr/K984	pr/K983	ba3817b5	docs	S5 docs/ chore	docs(smp): xpack-dev-smp.md Part II -- the commit-only migration procedure
126	pr/K985	pr/K984	6fb211a9	docs	S5 docs/ chore	docs(smp): xpack-dev-smp.md Part III -- execution record, step summary, commit list
127	pr/K986-STATUS	pr/K985	e2a90a42	docs	S5 docs/ chore	docs: STATUS for the commit- only migration of 2026-10-06
128	pr/K986-aarch32- second-board	pr/K986-STATUS	7bfb05be	docs	S5 docs/ chore	docs: aarch32-second-board -- the RK3506 SD driver is in the port
129	pr/K986- building- aarch32-aarch64	pr/K986-aarch32- second-board	a256bafe	docs	S5 docs/ chore	docs: building-aarch32-aarch64 without the devices repository
130	pr/K986-posix- arch-port	pr/K986- building- aarch32-aarch64	6c6a136a	docs	S5 docs/ chore	docs: posix-arch-port -- drivers and SD back-end are in the port
131	pr/K986-cortexm- port	pr/K986-posix- arch-port	a9c1ad47	docs	S5 docs/ chore	docs: cortexm-port -- the SoC directories and the verified results
132	pr/K986-test- smpl	pr/K986-cortexm- port	c273abb1	docs	S5 docs/ chore	docs: test-smpl lists the verdict contract and the no- power-cycle configs
133	pr/K986-tests- in-aarch32- aarch64	pr/K986-test- smpl	0ff357af	docs	S5 docs/ chore	docs: tests-in-aarch32-aarch64 shows drivers/ and soc/ in the port
134	pr/K986-WORK- SMP-HARNESS- GUIDE	pr/K986-tests- in-aarch32- aarch64	359b2b73	docs	S5 docs/ chore	docs(tests): WORK-SMP harness guide -- no devices repository
135	pr/K986-TESTS- XPACK-SYSTEM	pr/K986-WORK- SMP-HARNESS- GUIDE	982522d7	docs	S5 docs/ chore	docs(tests): TESTS-XPACK-SYSTEM -- the four ports, devices dissolved
136	pr/K986-HARNESS- TESTS-PARADIGM	pr/K986-TESTS- XPACK-SYSTEM	a8983cd6	docs	S5 docs/ chore	docs(tests): HARNESS-TESTS- PARADIGM -- no devices repository
137	pr/K986-HARNESS- BOARD-TEST- CHEATSHEET	pr/K986-HARNESS- TESTS-PARADIGM	ef3d8096	docs	S5 docs/ chore	docs(tests): HARNESS-BOARD- TEST-CHEATSHEET -- no devices repository
138	pr/K986-AARCH32- RPI-ZERO-2W- TESTS	pr/K986-HARNESS- BOARD-TEST- CHEATSHEET	997146ca	docs	S5 docs/ chore	docs(tests): AARCH32-RPI- ZERO-2W-TESTS -- drivers are in the port
139	pr/K986-STEPS	pr/K986-AARCH32- RPI-ZERO-2W- TESTS	335b3416	docs	S5 docs/ chore	docs(tests): STEPS -- a port finds only the kernel as a sibling
140	pr/K986-TESTS- DEVELOPER-GUIDE	pr/K986-STEPS	3ee23fe2	docs	S5 docs/ chore	docs(tests): TESTS-DEVELOPER- GUIDE -- the workspace without devices
141	pr/K986-TESTS- CATALOG	pr/K986-TESTS- DEVELOPER-GUIDE	63f2b822	docs	S5 docs/ chore	docs(tests): TESTS-CATALOG -- four ports, and the last verified run
142	pr/K986-tests- README-DEVELOPER	pr/K986-TESTS- CATALOG	d504cbc4	docs	S5 docs/ chore	docs(tests): README-DEVELOPER -- the current clone layout
143	pr/K986-tests- README- MAINTAINER	pr/K986-tests- README-DEVELOPER	0ddf9763	docs	S5 docs/ chore	docs(tests): README-MAINTAINER -- the current clone layout
144	pr/K986-test- framework	pr/K986-tests- README- MAINTAINER	b60858d1	docs	S5 docs/ chore	docs(tests): test-framework points to the smp side

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
145	pr/K986-agy-review	pr/K986-test-framework	8c3477d2	docs	S5 docs/ chore	docs(smp): agy-review -- status note, superseded by xpack-dev-smp.md
146	pr/K986-DeepSeek-review	pr/K986-agy-review	60e4d0ef	docs	S5 docs/ chore	docs(smp): DeepSeek-review -- status note, superseded by xpack-dev-smp.md
147	pr/K986-files-modif-by-step	pr/K986-DeepSeek-review	1da601cd	docs	S5 docs/ chore	docs(smp): files-modif-by-step -- status note, superseded by xpack-dev-smp.md
148	pr/K986-GITHUB-PROGRESSIVE-PR-GUIDE	pr/K986-files-modif-by-step	66ef2fa5	docs	S5 docs/ chore	docs(smp): GITHUB-PROGRESSIVE-PR-GUIDE -- status note, superseded by xpack-dev-smp.md
149	pr/K986-Implementation-SMP-Integration	pr/K986-GITHUB-PROGRESSIVE-PR-GUIDE	440716ba	docs	S5 docs/ chore	docs(smp): Implementation-SMP-Integration -- status note, superseded by xpack-dev-smp.md
150	pr/K986-micro-os-plus-iii-project-unification	pr/K986-Implementation-SMP-Integration	c6c7eebe	docs	S5 docs/ chore	docs(smp): project-unification -- status note
151	pr/K986-MICRO-OS-PLUS-SMP-VS-SINGLECORE-ANALYSIS	pr/K986-micro-os-plus-iii-project-unification	df0f0ed9	docs	S5 docs/ chore	docs(smp): SMP-VS-SINGLECORE-ANALYSIS -- status note
152	pr/K986-new-modifications	pr/K986-MICRO-OS-PLUS-SMP-VS-SINGLECORE-ANALYSIS	2be02079	docs	S5 docs/ chore	docs(smp): new-modifications -- status note
153	pr/K986-pull-request	pr/K986-new-modifications	0e1416d3	docs	S5 docs/ chore	docs(smp): pull-request -- status note, superseded by xpack-dev-smp.md
154	pr/K986-smp-construction	pr/K986-pull-request	02e62612	docs	S5 docs/ chore	docs(smp): smp-construction -- status note
155	pr/K986-SMP-UPSTREAM-INTEGRATION-PLAN	pr/K986-smp-construction	21d772ad	docs	S5 docs/ chore	docs(smp): SMP-UPSTREAM-INTEGRATION-PLAN -- status note, superseded by xpack-dev-smp.md
156	pr/K986-2026-09-20-micro-os-plus-iii-smp-unification-design	pr/K986-SMP-UPSTREAM-INTEGRATION-PLAN	039c42c6	docs	S5 docs/ chore	docs(specs): smp-unification-design -- status note
157	pr/K987	pr/K986-2026-09-20-micro-os-plus-iii-smp-unification-design	646a4904	docs	S5 docs/ chore	docs(smp): xpack-dev-smp.md lists the documentation updates
158	pr/K988	pr/K987	ec40386b	docs	S5 docs/ chore	docs(smp): xpack-dev-smp.md -- the pull requests of every repository
159	pr/K989	pr/K988	(this commit)	docs	S5 docs/ chore	docs(smp): xpack-dev-smp.md -- the verification replay from golden

micro-os-plus-iii-posix-arch -- 24 pull requests
(github.com/dan-maio/micro-os-plus-iii-posix-arch)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/P01	xpack-development	26a120bf	new code	S0 devices	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers
2	pr/P02	pr/P01	b8e8b4ad	new code	S0 devices	feat(soc): import the native host-file SD backend
3	pr/P03	pr/P02	9c5914ed	build	S0 devices	build(cmake): define the dissolved devices targets locally
4	pr/P04	pr/P03	5a1a255a	new code	S1 hrclock · G-hrclock	feat(port): clock_highres reads CLOCK_MONOTONIC
5	pr/P05	pr/P04	9be889da	new code	S2 host SMP	feat(port): the host is a multi-core machine -- one host thread per CPU
6	pr/P06	pr/P05	22f54df7	new code	S2 host SMP	feat(port): the board verdict contract on the host
7	pr/P20	pr/P06	e4f6c62f	build	S4 harness · G-harness	build(cmake): the port selects its board, and builds its own tests
8	pr/P21	pr/P20	305de44c	test	S4 test	test(native): rtos-apis at OS_NCPU = 1
9	pr/P22	pr/P21	e080ea6a	test	S4 test	test(native): mutex-stress
10	pr/P23	pr/P22	76c72864	test	S4 test	test(native): flatfs-test -- append to a file created empty
11	pr/P24	pr/P23	e648c634	test	S4 test	test(native): mutex-ceiling-test -- a refused protect lock leaves nothing behind
12	pr/P25	pr/P24	f0820dd5	test	S4 test	test(native): smp_test0 -- bring-up on one active CPU
13	pr/P26	pr/P25	c249804c	test	S4 test	test(native): smp_test1 -- semaphore ping-

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						pong across CPUs
14	pr/P27	pr/P26	4f1eee6b	test	S4 test	test(native): smp_test2 -- kernel lock coherency on every CPU
15	pr/P28	pr/P27	c26ff0fe	test	S4 test	test(native): smp_test3 -- message queue producer/consumer across CPUs
16	pr/P29	pr/P28	b67adcae	test	S4 test	test(native): smp_test4 -- load balancing without affinity
17	pr/P30	pr/P29	75a487c7	test	S4 test	test(native): smp-num-test -- concurrent workers and console I/O
18	pr/P31	pr/P30	9fe414ca	test	S4 test	test(native): smp-mat-test -- parallel block solver (N=500, B=50)
19	pr/P32	pr/P31	e60f2f2c	test	S4 test	test(native): smp-pipeline-test -- multi-stage pipeline across CPUs
20	pr/P33	pr/P32	e614e825	test	S4 test	test(native): smp-pro-cons-test -- producers and consumers over every kernel object
21	pr/P34	pr/P33	c4f5f652	test	S4 test	test(native): smp-mutex-stress -- the SMP leg of mutex-stress, with per-core checks
22	pr/P35	pr/P34	6aec9d4c	test	S4 test	test(native): smp-rtos-apis -- the SMP leg of rtos-apis
23	pr/P14	pr/P35	e156e975	tool	tool	tools: a probe showing why TSan fiber annotations cannot work here
24	pr/P36	pr/P14	93c42239	chore	chore	chore: ignore every build* directory and the build products

micro-os-plus-iii-cortexm -- 86 pull requests
(github.com/dan-maio/micro-os-plus-iii-cortexm)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/C01	xpack-development	b2bdd12c	new code	S0 devices	feat(soc): import the STM32F4xx SoC support
2	pr/C02	pr/C01	2310c355	new code	S0 devices	feat(soc): import the RP2350 SoC support
3	pr/C03	pr/C02	f64c09a4	build	S0 devices	build(cmake): define the dissolved SoC targets locally
4	pr/C04	pr/C03	736010e8	new code	S1 hrclock · G-hrclock	feat(port): clock_highres declares it has no hardware counter
5	pr/C04b	pr/C04	31b04065	correction	S1 hrclock	fix(port): cycles_since_tick() reads the SysTick pending flag from ICSR
6	pr/C05	pr/C04b	dfbfd56	new code	S2 Cortex-M SMP	feat(port): the SMP port contract on the generic Cortex-M core
7	pr/C06	pr/C05	cc6412ab	new code	S2 Cortex-M SMP	feat(port): the generic Cortex-M core runs the SMP scheduler on one CPU
8	pr/C07	pr/C06	c9b5c630	new code	S2 Cortex-M SMP	feat(libc): a _getentropy() stub for the QEMU images
9	pr/C08	pr/C07	597bdda8	new code	S2 Cortex-M SMP	feat: _Exit() reports the test verdict through semihosting
10	pr/C09	pr/C08	ba9ef5ee	new code	S3 new silicon	feat(port): a generic Cortex-M33 (SSE-200) dual-core port for QEMU
11	pr/C10	pr/C09	0071da04	new code	S3 new silicon	feat(port): the RP2350 dual-core port
12	pr/C20a	pr/C10	591b070e	test	S4 board/harness	test(nucleof411): the NUCLE0-F411RE board
13	pr/C20b	pr/C20a	3629ead6	build	S4 board/harness · G-harness	build(cmake): the port selects its board, builds its tests, and exports the QEMU cores
14	pr/C21	pr/C20b	2187bd34	test	S4 board/harness	test(pico2): the Raspberry Pi Pico 2 board (RP2350, two Cortex-M33)
15	pr/C21b	pr/C21	fbd44d24	test	S4 board/harness	test(pico2): vendor the TinyUSB device stack for the RP2350
16	pr/C21c	pr/C21b	e82f3f7e	test		

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
					S4 board/harness	test(pico2): a minimal pico-sdk shim for TinyUSB
17	pr/C21d	pr/C21c	9e62cb6b	test	S4 board/harness	test(pico2): CDC-ACM and HID device configurations
18	pr/C22-rtos-apis	pr/C21d	13e72795	test	S4 test	test(pico2): rtos-apis
19	pr/C22-cmsis-os-validator-ram	pr/C22-rtos-apis	094d7daf	test	S4 test	test(pico2): cmsis-os-validator-ram
20	pr/C22-cmsis-os-validator	pr/C22-cmsis-os-validator-ram	e58e7663	test	S4 test	test(pico2): cmsis-os-validator
21	pr/C22-exc-test	pr/C22-cmsis-os-validator	49e47089	test	S4 test	test(pico2): exc-test
22	pr/C22-fp-switch	pr/C22-exc-test	3cbb8aff	test	S4 test	test(pico2): fp-switch
23	pr/C22-mutex-stress-ram	pr/C22-fp-switch	88bd02ad	test	S4 test	test(pico2): mutex-stress-ram
24	pr/C22-mutex-stress	pr/C22-mutex-stress-ram	c8fbb351	test	S4 test	test(pico2): mutex-stress
25	pr/C22-rtos-apis-ram	pr/C22-mutex-stress	0a2e9b77	test	S4 test	test(pico2): rtos-apis-ram
26	pr/C22-sc-test-ko	pr/C22-rtos-apis-ram	89a5454d	test	S4 test	test(pico2): sc-test-ko
27	pr/C22-smp-mat-test	pr/C22-sc-test-ko	57149d31	test	S4 test	test(pico2): smp-mat-test
28	pr/C22-smp-mat-test-ram	pr/C22-smp-mat-test	d3bd1bff	test	S4 test	test(pico2): smp-mat-test-ram
29	pr/C22-smp-test-ko	pr/C22-smp-mat-test-ram	d4fb4e67	test	S4 test	test(pico2): smp-test-ko
30	pr/C22-smp-test-usb-cdc-acm	pr/C22-smp-test-ko	6156aa38	test	S4 test	test(pico2): smp-test-usb-cdc-acm
31	pr/C22-smp-test-usb-hid	pr/C22-smp-test-usb-cdc-acm	385cb24c	test	S4 test	test(pico2): smp-test-usb-hid
32	pr/C22-smp-test0	pr/C22-smp-test-usb-hid	e7546968	test	S4 test	test(pico2): smp-test0
33	pr/C22-smp-test1	pr/C22-smp-test0	a8263f0d	test	S4 test	test(pico2): smp-test1
34	pr/C22-smp-test2	pr/C22-smp-test1	137a1cb1	test	S4 test	test(pico2): smp-test2
35			802c05fd	test	S4 test	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
	pr/C22-smp-test3	pr/C22-smp-test2				test(pico2): smp-test3
36	pr/C22-smp-test4	pr/C22-smp-test3	30d31e49	test	S4 test	test(pico2): smp-test4
37	pr/C22-smp-test5	pr/C22-smp-test4	9367c2fa	test	S4 test	test(pico2): smp-test5
38	pr/C23	pr/C22-smp-test5	1cffd384	test	S4 board/harness	test(pico2-rp2350b-psram): the RP2350B board with 16 MiB flash and PSRAM
39	pr/C23-exc-test	pr/C23	ecbfb0b0	test	S4 test	test(pico2-rp2350b-psram): exc-test
40	pr/C23-fp-switch	pr/C23-exc-test	f2708769	test	S4 test	test(pico2-rp2350b-psram): fp-switch
41	pr/C23-sc-test-ko	pr/C23-fp-switch	9767e0dc	test	S4 test	test(pico2-rp2350b-psram): sc-test-ko
42	pr/C23-smp-mat-test	pr/C23-sc-test-ko	5a78743c	test	S4 test	test(pico2-rp2350b-psram): smp-mat-test
43	pr/C23-smp-test-ko	pr/C23-smp-mat-test	eb888556	test	S4 test	test(pico2-rp2350b-psram): smp-test-ko
44	pr/C23-smp-test-nested-clock	pr/C23-smp-test-ko	1c50c6c1	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested-clock
45	pr/C23-smp-test-nested-clock_200	pr/C23-smp-test-nested-clock	dcc8fdf3	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested-clock_200
46	pr/C23-smp-test-nested-clock_250	pr/C23-smp-test-nested-clock_200	53e7990b	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested-clock_250
47	pr/C23-smp-test-nested	pr/C23-smp-test-nested-clock_250	03462dd0	test	S4 test	test(pico2-rp2350b-psram): smp-test-nested
48	pr/C23-smp-test0	pr/C23-smp-test-nested	4fddd1a6	test	S4 test	test(pico2-rp2350b-psram): smp-test0
49	pr/C23-smp-test1	pr/C23-smp-test0	38f862a7	test	S4 test	test(pico2-rp2350b-psram): smp-test1
50	pr/C23-smp-test2	pr/C23-smp-test1	792ee80b	test	S4 test	test(pico2-rp2350b-psram): smp-test2
51	pr/C23-smp-test3	pr/C23-smp-test2	fdcd879c	test	S4 test	test(pico2-rp2350b-psram): smp-test3
52	pr/C23-smp-test4	pr/C23-smp-test3	0b2d5838	test	S4 test	test(pico2-rp2350b-psram): smp-test4
53	pr/C23-smp-test5	pr/C23-smp-test4	15867310	test	S4 test	test(pico2-rp2350b-psram): smp-test5
54	pr/C24	pr/C23-smp-test5	994cb864	test	S4 board/harness	test(pico2-pizero): the RP2350 on a Pi Zero form-factor carrier
55	pr/C24-exc-test	pr/C24	e450aaff	test	S4 test	test(pico2-pizero): exc-test

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
56	pr/C24-psram-exec	pr/C24-exc-test	ab90bf91	test	S4 test	test(pico2-pizero): psram-exec
57	pr/C24-psram-mat-test-250	pr/C24-psram-exec	76a2d22f	test	S4 test	test(pico2-pizero): psram-mat-test-250
58	pr/C24-sc-test-ko	pr/C24-psram-mat-test-250	25cb812e	test	S4 test	test(pico2-pizero): sc-test-ko
59	pr/C24-smp-mat-test	pr/C24-sc-test-ko	43d09f78	test	S4 test	test(pico2-pizero): smp-mat-test
60	pr/C24-smp-test-ko	pr/C24-smp-mat-test	395b0d10	test	S4 test	test(pico2-pizero): smp-test-ko
61	pr/C24-smp-test-usb-cdc-acm	pr/C24-smp-test-ko	038509f6	test	S4 test	test(pico2-pizero): smp-test-usb-cdc-acm
62	pr/C24-smp-test-usb-hid	pr/C24-smp-test-usb-cdc-acm	1eea852d	test	S4 test	test(pico2-pizero): smp-test-usb-hid
63	pr/C24-smp-test0	pr/C24-smp-test-usb-hid	044c5bde	test	S4 test	test(pico2-pizero): smp-test0
64	pr/C24-smp-test1	pr/C24-smp-test0	12570a64	test	S4 test	test(pico2-pizero): smp-test1
65	pr/C24-smp-test2	pr/C24-smp-test1	924b36a0	test	S4 test	test(pico2-pizero): smp-test2
66	pr/C24-smp-test3	pr/C24-smp-test2	807afed1	test	S4 test	test(pico2-pizero): smp-test3
67	pr/C24-smp-test4	pr/C24-smp-test3	cb72fb3e	test	S4 test	test(pico2-pizero): smp-test4
68	pr/C24-smp-test5	pr/C24-smp-test4	b6459bcb	test	S4 test	test(pico2-pizero): smp-test5
69	pr/C25-cmsis-os-validator	pr/C24-smp-test5	14bba6b7	test	S4 test	test(nucleof411): cmsis-os-validator
70	pr/C25-mos-test1	pr/C25-cmsis-os-validator	82c09863	test	S4 test	test(nucleof411): mos-test1
71	pr/C25-mutex-stress	pr/C25-mos-test1	5a3c8ec6	test	S4 test	test(nucleof411): mutex-stress
72	pr/C25-rtos-apis	pr/C25-mutex-stress	904bc935	test	S4 test	test(nucleof411): rtos-apis
73	pr/C26	pr/C25-rtos-apis	b695919a	test	S4 board/harness	test(weactf411): the WeAct Black Pill F411 board
74	pr/C26-cmsis-os-validator	pr/C26	3edc06c0	test	S4 test	test(weactf411): cmsis-os-validator
75	pr/C26-mos-test1	pr/C26-cmsis-os-validator	900c9402	test	S4 test	test(weactf411): mos-test1
76			7b9526bb	test	S4 test	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
	pr/C26-mutex-stress	pr/C26-mos-test1				test(weactf411): mutex-stress
77	pr/C26-rtos-apis	pr/C26-mutex-stress	c8d67d58	test	S4 test	test(weactf411): rtos-apis
78	pr/C26-spi-pipeline	pr/C26-rtos-apis	6935c185	test	S4 test	test(weactf411): spi-pipeline
79	pr/C27	pr/C26-spi-pipeline	dc22fcdb	test	S4 board/harness	test(weactf412): the WeAct F412 board
80	pr/C27-cmsis-os-validator	pr/C27	a5f805b5	test	S4 test	test(weactf412): cmsis-os-validator
81	pr/C27-mos-test1	pr/C27-cmsis-os-validator	f2f0695c	test	S4 test	test(weactf412): mos-test1
82	pr/C27-mutex-stress	pr/C27-mos-test1	efa2b9a7	test	S4 test	test(weactf412): mutex-stress
83	pr/C27-rtos-apis	pr/C27-mutex-stress	bc32b726	test	S4 test	test(weactf412): rtos-apis
84	pr/C27-uart-test1	pr/C27-rtos-apis	efd8432c	test	S4 test	test(weactf412): uart-test1
85	pr/C28	pr/C27-uart-test1	72149579	chore	S5 docs/chore	chore: ignore every build* directory, the build products and the USB host tool
86	pr/C29	pr/C28	c5a71fbc	docs	S5 docs/chore	docs: README for the six boards and the generic QEMU cores

micro-os-plus-iii-aarch32 -- 64 pull requests
(github.com/dan-maio/micro-os-plus-iii-aarch32)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/A32-01	xpack-development	03a0cfc1	new code	S0 devices	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers
2	pr/A32-02	pr/A32-01	c318b741	new code	S0 devices	feat(soc): import the BCM2837 (Raspberry Pi 3 B / Zero 2 W) SoC support
3	pr/A32-03	pr/A32-02	58cff47a	new code	S0 devices	feat(soc): import the RK3506 (Luckfox Lyra) SoC support
4	pr/A32-04	pr/A32-03	a8662d6e	chore	S3 port	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
						chore: ignore every build* directory
5	pr/A32-05	pr/A32-04	b0eb0ed7	new code	S3 port	feat(port): AArch32 port declarations, application config and device header
6	pr/A32-06	pr/A32-05	81d97c4d	new code	S3 port	feat(port): ARMv7-A MMU bring-up
7	pr/A32-07	pr/A32-06	583aa440	new code	S3 port	feat(port): AArch32 exception vectors, IRQ dispatch and fault dump
8	pr/A32-08	pr/A32-07	4d9e3c1b	new code	S3 port	feat(port): ARM generic timer, and the high-resolution counter from CNTPCT
9	pr/A32-09	pr/A32-08	e510bd9f	new code	S3 port	feat(port): AArch32 context switch and the port half of the SMP scheduler
10	pr/A32-10	pr/A32-09	37b944cf	new code	S3 port	feat(port): secondary-core entry
11	pr/A32-11	pr/A32-10	6eb8c2aa	new code	S3 port	feat(port): semihosting console and a strong _Exit()
12	pr/A32-20	pr/A32-11	8cac0c56	test	S4 board/ harness	test(rpi-zero-2w): the Raspberry Pi Zero 2 W board (BCM2837, 4x Cortex-A53, AArch32)
13	pr/A32-21	pr/A32-20	576597f0	build	S4 board/ harness	build(cmake): the AArch32 port project, its board model and its test builder
14	pr/A32-T-smp_test0	pr/A32-21	0308bd45	test	S4 test	test(rpi-zero-2w): smp_test0
15	pr/A32-T-smp-mat-test	pr/A32-T-smp_test0	3f8b9a64	test	S4 test	test(rpi-zero-2w): smp-mat-test
16	pr/A32-T-cmsis-os-validator	pr/A32-T-smp-mat-test	8bccb4ee	test	S4 test	test(rpi-zero-2w): cmsis-os-validator
17			cc8d43bb	test	S4 test	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
	pr/A32-T-mutex-stress	pr/A32-T-cmsis-os-validator				test(rpi-zero-2w): mutex-stress
18	pr/A32-T-rtos-apis	pr/A32-T-mutex-stress	f53d5e9e	test	S4 test	test(rpi-zero-2w): rtos-apis
19	pr/A32-T-sd_test	pr/A32-T-rtos-apis	286da73d	test	S4 test	test(rpi-zero-2w): sd_test
20	pr/A32-T-smp-mat-sdcard-test	pr/A32-T-sd_test	2bd8db32	test	S4 test	test(rpi-zero-2w): smp-mat-sdcard-test
21	pr/A32-T-smp-num-test	pr/A32-T-smp-mat-sdcard-test	4127ec3c	test	S4 test	test(rpi-zero-2w): smp-num-test
22	pr/A32-T-smp-pipeline-test	pr/A32-T-smp-num-test	89a1b78c	test	S4 test	test(rpi-zero-2w): smp-pipeline-test
23	pr/A32-T-smp-pro-cons-test	pr/A32-T-smp-pipeline-test	87ca7814	test	S4 test	test(rpi-zero-2w): smp-pro-cons-test
24	pr/A32-T-smp_test1	pr/A32-T-smp-pro-cons-test	98efc7ea	test	S4 test	test(rpi-zero-2w): smp_test1
25	pr/A32-T-smp_test2	pr/A32-T-smp_test1	3f4148c8	test	S4 test	test(rpi-zero-2w): smp_test2
26	pr/A32-T-smp_test3	pr/A32-T-smp_test2	d4443aca	test	S4 test	test(rpi-zero-2w): smp_test3
27	pr/A32-T-smp_test4	pr/A32-T-smp_test3	bd38979a	test	S4 test	test(rpi-zero-2w): smp_test4
28	pr/A32-T-usb_test	pr/A32-T-smp_test4	aa7fae53	test	S4 test	test(rpi-zero-2w): usb_test
29	pr/A32-R3	pr/A32-T-usb_test	aa0b83c0	test	S4 board/harness	test(rpi3b): the Raspberry Pi 3 B board (AArch32), sharing the Zero 2 W sources
30	pr/A32-R3-smp_test0	pr/A32-R3	22010a2c	test	S4 test	test(rpi3b): smp_test0
31	pr/A32-R3-cmsis-os-validator	pr/A32-R3-smp_test0	e0df39c7	test	S4 test	test(rpi3b): cmsis-os-validator
32	pr/A32-R3-mutex-stress	pr/A32-R3-cmsis-os-validator	b3666f39	test	S4 test	test(rpi3b): mutex-stress
33	pr/A32-R3-rtos-apis	pr/A32-R3-mutex-stress	832c71ae	test	S4 test	test(rpi3b): rtos-apis
34	pr/A32-R3-sd_test	pr/A32-R3-rtos-apis	8de31552	test	S4 test	test(rpi3b): sd_test
35	pr/A32-R3-smp-mat-sdcard-test	pr/A32-R3-sd_test	465e7d4a	test	S4 test	test(rpi3b): smp-mat-sdcard-test

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
36	pr/A32-R3-smp-mat-test	pr/A32-R3-smp-mat-sdcard-test	c80191d9	test	S4 test	test(rpi3b): smp-mat-test
37	pr/A32-R3-smp-num-test	pr/A32-R3-smp-mat-test	4359e4ce	test	S4 test	test(rpi3b): smp-num-test
38	pr/A32-R3-smp-pipeline-test	pr/A32-R3-smp-num-test	385f0938	test	S4 test	test(rpi3b): smp-pipeline-test
39	pr/A32-R3-smp-pro-cons-test	pr/A32-R3-smp-pipeline-test	b5235187	test	S4 test	test(rpi3b): smp-pro-cons-test
40	pr/A32-R3-smp_test1	pr/A32-R3-smp-pro-cons-test	436d38c9	test	S4 test	test(rpi3b): smp_test1
41	pr/A32-R3-smp_test2	pr/A32-R3-smp_test1	a121a4ca	test	S4 test	test(rpi3b): smp_test2
42	pr/A32-R3-smp_test3	pr/A32-R3-smp_test2	6b56b17f	test	S4 test	test(rpi3b): smp_test3
43	pr/A32-R3-smp_test4	pr/A32-R3-smp_test3	bd4569eb	test	S4 test	test(rpi3b): smp_test4
44	pr/A32-R3-usb_test	pr/A32-R3-smp_test4	a33a5cc0	test	S4 test	test(rpi3b): usb_test
45	pr/A32-LL	pr/A32-R3-usb_test	b468cbba	test	S4 board/harness	test(luckfox-lyra): the Luckfox Lyra B board (RK3506, 3x Cortex-A7, hardware only)
46	pr/A32-LL-smp_test0	pr/A32-LL	e617e017	test	S4 test	test(luckfox-lyra): smp_test0
47	pr/A32-LL-smp_test_int5	pr/A32-LL-smp_test0	067f4a19	test	S4 test	test(luckfox-lyra): smp_test_int5
48	pr/A32-LL-sd_test	pr/A32-LL-smp_test_int5	79eca794	test	S4 test	test(luckfox-lyra): sd_test
49	pr/A32-LL-smp-mat-sdcard-test	pr/A32-LL-sd_test	aa5a77ac	test	S4 test	test(luckfox-lyra): smp-mat-sdcard-test
50	pr/A32-LL-smp-mat-test	pr/A32-LL-smp-mat-sdcard-test	fcaeda34	test	S4 test	test(luckfox-lyra): smp-mat-test
51	pr/A32-LL-smp-num-test	pr/A32-LL-smp-mat-test	3f2f7a29	test	S4 test	test(luckfox-lyra): smp-num-test
52	pr/A32-LL-smp-pipeline-test	pr/A32-LL-smp-num-test	7c9a0205	test	S4 test	test(luckfox-lyra): smp-pipeline-test
53	pr/A32-LL-smp-pro-cons-test	pr/A32-LL-smp-pipeline-test	b42eaf86	test	S4 test	test(luckfox-lyra): smp-pro-cons-test
54	pr/A32-LL-smp_test1	pr/A32-LL-smp-pro-cons-test	bd934140	test	S4 test	test(luckfox-lyra): smp_test1
55	pr/A32-LL-smp_test2	pr/A32-LL-smp_test1	09b7ce26	test	S4 test	test(luckfox-lyra): smp_test2

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
56	pr/A32-LL-smp_test3	pr/A32-LL-smp_test2	0b58a238	test	S4 test	test(luckfox-lyra): smp_test3
57	pr/A32-LL-smp_test4	pr/A32-LL-smp_test3	9a0a34a0	test	S4 test	test(luckfox-lyra): smp_test4
58	pr/A32-LL-smp_test5	pr/A32-LL-smp_test4	979865e4	test	S4 test	test(luckfox-lyra): smp_test5
59	pr/A32-LL-smp_test6	pr/A32-LL-smp_test5	1cdf9eb1	test	S4 test	test(luckfox-lyra): smp_test6
60	pr/A32-LL-smp_test7	pr/A32-LL-smp_test6	0a1f7b1e	test	S4 test	test(luckfox-lyra): smp_test7
61	pr/A32-LL-smp_test_int	pr/A32-LL-smp_test7	3ce4a94a	test	S4 test	test(luckfox-lyra): smp_test_int
62	pr/A32-LL-smp_test_int2	pr/A32-LL-smp_test_int	d3f74a0f	test	S4 test	test(luckfox-lyra): smp_test_int2
63	pr/A32-LL-smp_test_int3	pr/A32-LL-smp_test_int2	d254ad93	test	S4 test	test(luckfox-lyra): smp_test_int3
64	pr/A32-LL-smp_test_int4	pr/A32-LL-smp_test_int3	0aa86aed	test	S4 test	test(luckfox-lyra): smp_test_int4

micro-os-plus-iii-aarch64 -- 43 pull requests
(github.com/dan-maio/micro-os-plus-iii-aarch64)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/A64-01	xpack-development	b764bc98	new code	S0 devices	feat(drivers): import the neutral SD, flatfs, USB and FatFs drivers
2	pr/A64-02	pr/A64-01	fe04ec15	new code	S0 devices	feat(soc): import the BCM2837 (Raspberry Pi 3 B / Zero 2 W) SoC support
3	pr/A64-03	pr/A64-02	30822be2	chore	S3 port	chore: ignore every build* directory
4	pr/A64-04	pr/A64-03	f04d5ef5	new code	S3 port	feat(port): AArch64 port declarations, application config and device header
5	pr/A64-05	pr/A64-04	522cf173	new code	S3 port	feat(port): AArch64 MMU bring-up

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
6	pr/A64-06	pr/A64-05	265ce322	new code	S3 port	feat(port): AArch64 exception reporting and the static constructor runners
7	pr/A64-07	pr/A64-06	f5359a25	new code	S3 port	feat(port): ARM generic timer accessors (CNTFRQ_EL0, CNTPCT_EL0)
8	pr/A64-08	pr/A64-07	e04efa95	new code	S3 port	feat(port): AArch64 context frame and the port half of the SMP scheduler
9	pr/A64-09	pr/A64-08	ba24acc6	new code	S3 port	feat(port): secondary-core entry
10	pr/A64-10	pr/A64-09	7e796801	new code	S3 port	feat(port): semihosting console
11	pr/A64-20	pr/A64-10	9cafe2de	test	S4 board/harness	test(rpi-zero-2w): the Raspberry Pi Zero 2 W board (BCM2837, 4x Cortex-A53, AArch64)
12	pr/A64-21	pr/A64-20	475d0604	build	S4 board/harness	build(cmake): the AArch64 port project, its board model and its test builder
13	pr/A64-T-smp_test0	pr/A64-21	cca66872	test	S4 test	test(rpi-zero-2w): smp_test0
14	pr/A64-T-smp-mat-test	pr/A64-T-smp_test0	07d27cd6	test	S4 test	test(rpi-zero-2w): smp-mat-test
15	pr/A64-T-cmsis-os-validator	pr/A64-T-smp-mat-test	bc32bcd8	test	S4 test	test(rpi-zero-2w): cmsis-os-validator
16	pr/A64-T-mutex-stress	pr/A64-T-cmsis-os-validator	d87cebb7	test	S4 test	test(rpi-zero-2w): mutex-stress
17	pr/A64-T-rtos-apis	pr/A64-T-mutex-stress	99c5a5bf	test	S4 test	test(rpi-zero-2w): rtos-apis
18	pr/A64-T-sd_test	pr/A64-T-rtos-apis	d666d390	test	S4 test	test(rpi-zero-2w): sd_test
19	pr/A64-T-smp-mat-sdcard-test	pr/A64-T-sd_test	1731d0b1	test	S4 test	test(rpi-zero-2w): smp-mat-sdcard-test

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
20	pr/A64-T-smp-num-test	pr/A64-T-smp-mat-sdcard-test	edf06551	test	S4 test	test(rpi-zero-2w): smp-num-test
21	pr/A64-T-smp-pipeline-test	pr/A64-T-smp-num-test	8750df6e	test	S4 test	test(rpi-zero-2w): smp-pipeline-test
22	pr/A64-T-smp-pro-cons-test	pr/A64-T-smp-pipeline-test	5e0e5a65	test	S4 test	test(rpi-zero-2w): smp-pro-cons-test
23	pr/A64-T-smp_test1	pr/A64-T-smp-pro-cons-test	ed02a26b	test	S4 test	test(rpi-zero-2w): smp_test1
24	pr/A64-T-smp_test2	pr/A64-T-smp_test1	12c23d59	test	S4 test	test(rpi-zero-2w): smp_test2
25	pr/A64-T-smp_test3	pr/A64-T-smp_test2	216918dd	test	S4 test	test(rpi-zero-2w): smp_test3
26	pr/A64-T-smp_test4	pr/A64-T-smp_test3	87d619e3	test	S4 test	test(rpi-zero-2w): smp_test4
27	pr/A64-T-usb_test	pr/A64-T-smp_test4	5da24f9a	test	S4 test	test(rpi-zero-2w): usb_test
28	pr/A64-R3	pr/A64-T-usb_test	c84b8e00	test	S4 board/harness	test(rpi3b): the Raspberry Pi 3 B board (AArch64), sharing the Zero 2 W sources
29	pr/A64-R3-smp_test0	pr/A64-R3	b3fc5a12	test	S4 test	test(rpi3b): smp_test0
30	pr/A64-R3-cmsis-os-validator	pr/A64-R3-smp_test0	031d8f52	test	S4 test	test(rpi3b): cmsis-os-validator
31	pr/A64-R3-mutex-stress	pr/A64-R3-cmsis-os-validator	3ae3d571	test	S4 test	test(rpi3b): mutex-stress
32	pr/A64-R3-rtos-apis	pr/A64-R3-mutex-stress	e7bc6bc7	test	S4 test	test(rpi3b): rtos-apis
33	pr/A64-R3-sd_test	pr/A64-R3-rtos-apis	3e9f6e5a	test	S4 test	test(rpi3b): sd_test
34	pr/A64-R3-smp-mat-sdcard-test	pr/A64-R3-sd_test	ea725894	test	S4 test	test(rpi3b): smp-mat-sdcard-test
35	pr/A64-R3-smp-mat-test	pr/A64-R3-smp-mat-sdcard-test	0cce173e	test	S4 test	test(rpi3b): smp-mat-test
36	pr/A64-R3-smp-num-test	pr/A64-R3-smp-mat-test	d4f629c3	test	S4 test	test(rpi3b): smp-num-test
37	pr/A64-R3-smp-		a8082ca9	test	S4 test	

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
	pipeline-test	pr/A64-R3-smp-num-test				test(rpi3b): smp-pipeline-test
38	pr/A64-R3-smp-pro-cons-test	pr/A64-R3-smp-pipeline-test	1ace669b	test	S4 test	test(rpi3b): smp-pro-cons-test
39	pr/A64-R3-smp_test1	pr/A64-R3-smp-pro-cons-test	84bceba9	test	S4 test	test(rpi3b): smp_test1
40	pr/A64-R3-smp_test2	pr/A64-R3-smp_test1	c045d19b	test	S4 test	test(rpi3b): smp_test2
41	pr/A64-R3-smp_test3	pr/A64-R3-smp_test2	fe5a5489	test	S4 test	test(rpi3b): smp_test3
42	pr/A64-R3-smp_test4	pr/A64-R3-smp_test3	ebf7dce3	test	S4 test	test(rpi3b): smp_test4
43	pr/A64-R3-usb_test	pr/A64-R3-smp_test4	a723ecc1	test	S4 test	test(rpi3b): usb_test

micro-os-plus-iii-devices -- 1 pull request (github.com/dan-maio/micro-os-plus-iii-devices)

PR	Head branch (Commit-ID)	Base branch	SHA	Category	Stage / theme	Subject
1	pr/D01	smp	(bundle)	docs	S0 devices	docs: retire the repository -- its contents moved to the architecture ports

D01 is kept in devices-retired.bundle (the local clone was deleted after S0, §14.2). After it is merged, the repository is archived on GitHub.

22.4 Documentation updates after the migration (kernel PRs 126-159)

Made after the full test run, so that every document in *docs/* describes the executed state: Part III itself (K985); one commit per document (K986-*), which corrects stale facts in the reference documents (no *micro-os-plus-iii-devices*, no *UOS_DEVICES_DIR*, the current clone layout and the verified results) and adds a *Status 2026-10-06* note to the historical plans and reviews (their PDFs regenerated); the two commits that complete this section (K987, K988); and the verification record of §21.6 (K989). The upstream documents that do not concern the migration (*HISTORY*, *NOTES*, *TODO*, *posix-io**, *other-posix-systems*, *upstream**) are unchanged.